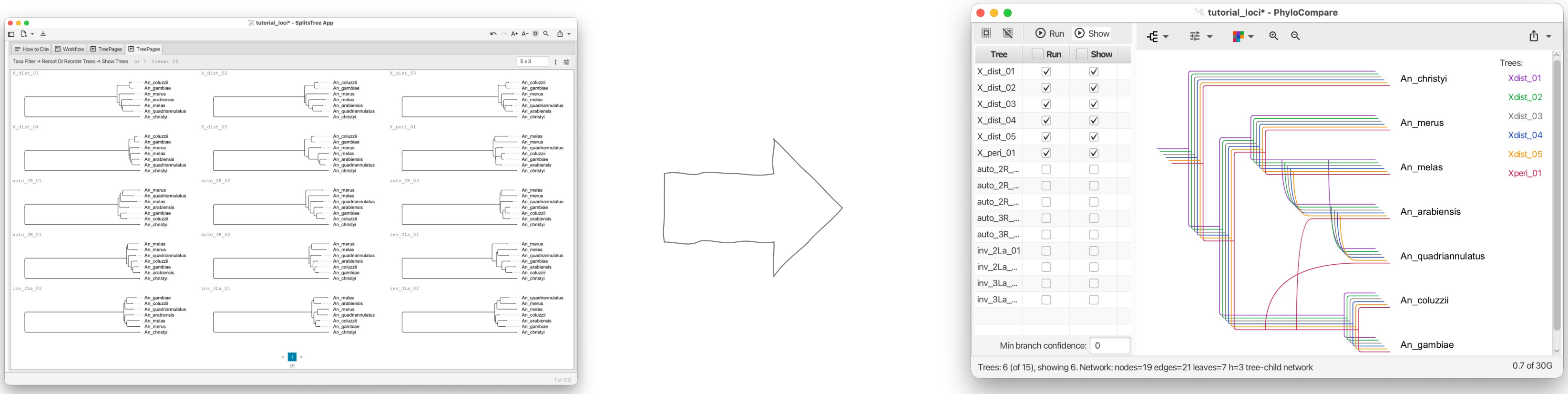
From Trees to Networks



ISMB 2026 Tutorial IP3

Daniel H. Huson, Anupam Gautam and Banu Cetinkaya

Version July 11, 2026

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Your presenters today

Daniel Huson

- Professor of Algorithms in Bioinformatics
- University of Tübingen
- Phylogenetic networks, SplitsTree, PhyloSketch, PhyloParallelograms

Anupam Gautam

- Post-Doc
- University of Tübingen
- Max-Planck Institute for Biology Tübingen
- PhyloGuide, AI-assisted phylogenetic analysis

Banu Cetinkaya

- PhD student
- University of Tübingen
- Phylogenetic networks, co-developer of PhyloParallelograms

What we'll do today (09:00 to 13:00)

09:00 - 10:50 Trees

- Per-locus inference with IQ-TREE and BEAST X
- Coalescent species tree with ASTRAL

➢ "We have many gene trees and they disagree"

10:50 - 11:10 / Break ☕

11:10 - 13:00 Networks

- SplitsTree & Neighbor-Net on alignments
- PhyloSketch for interactive network construction
- PhyloParallelograms for networks from gene tree sets

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1. Tutorial overview and dataset
2. Setup verification
3. **Multi-locus phylogenomics lecture**
4. IQ-TREE 3 hands-on
5. BEAST X + SplitsTree posterior visualization
6. ASTRAL hands-on

☕

7. **Networks lecture**
8. SplitsTree Neighbor-Net hands-on
9. PhyloSketch hands-on
10. PhyloParallelograms hands-on
11. Biological interpretation + wrap-up

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Dataset: the *Anopheles gambiae* complex

RESEARCH ARTICLE

2 JANUARY 2015

MOSQUITO GENOMICS

Extensive introgression in a malaria vector species complex revealed by phylogenomics

Michael C. Fontaine,[1,2*†] James B. Pease,[3*] Aaron Steele,[4] Robert M. Waterhouse,[5,6,7,8] Daniel E. Neafsey,[6] Igor V. Sharakhov,[9,10] Xiaofang Jiang,[10] Andrew B. Hall,[10] Flaminia Catteruccia,[11,12] Evdoxia Kakani,[11,12] Sara N. Mitchell,[11] Yi-Chieh Wu,[5] Hilary A. Smith,[1,2] R. Rebecca Love,[1,2] Mara K. Lawniczak,[13‡] Michel A. Slotman,[14] Scott J. Emrich,[2,4] Matthew W. Hahn,[3,15§] Nora J. Besansky[1,2§]

Science **347** (6217), 1258524. DOI: 10.1126/science.1258524

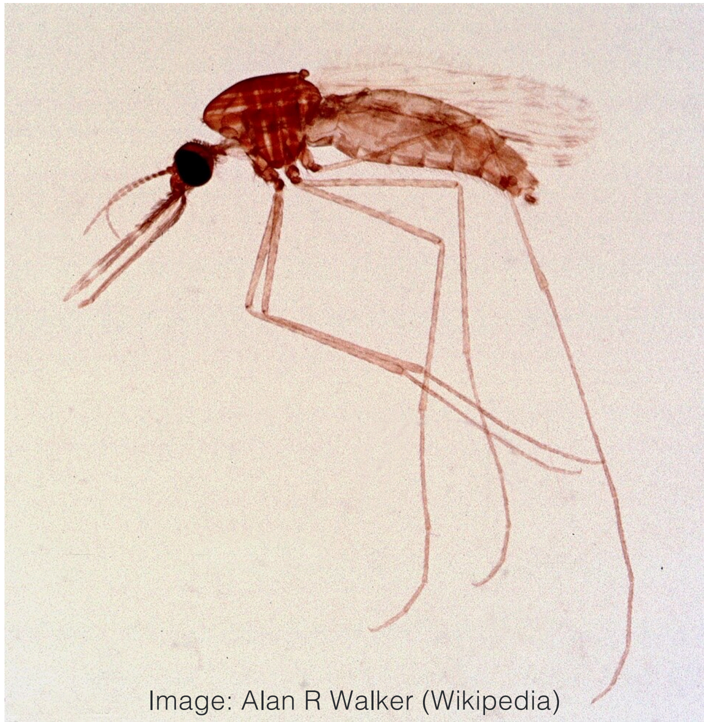
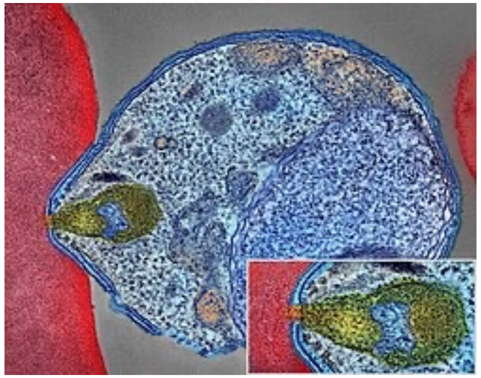


Image: Alan R Walker (Wikipedia)

Dataset: the *Anopheles gambiae* complex

Why public health cares:

- Three of six species are the principal vectors of *P. falciparum* malaria
- How did they acquire vectorial capacity?
- Can phylogeny help understand the disease



Malaria parasite connecting to a red blood cell
NIAID, Wikipedia

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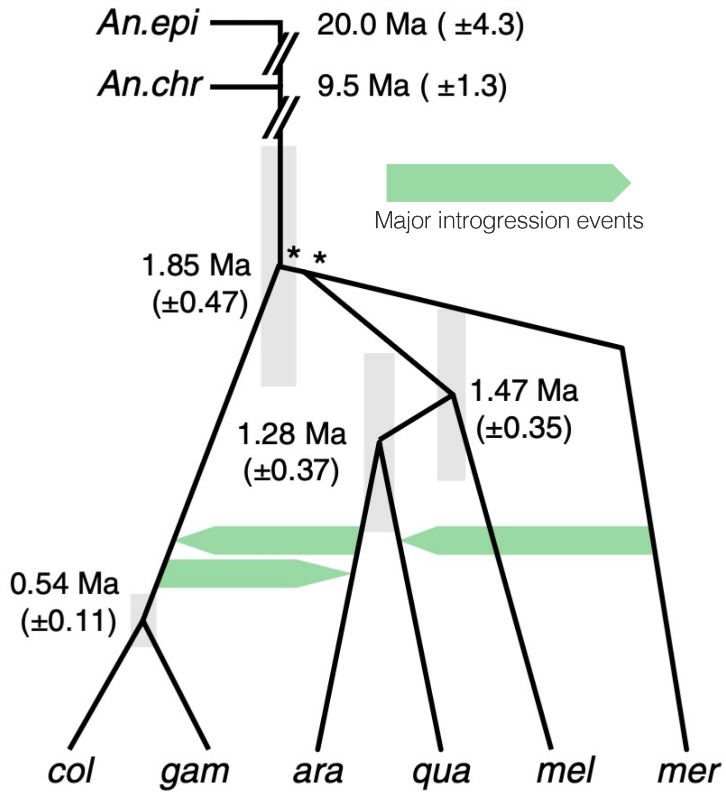
Dataset: the *Anopheles gambiae* complex

Why phylogenetics cares:

- Six closely related species, recently diverged (about 2 Mya)
- Pervasive autosomal introgression
- X chromosome tells a different story than autosomes
- Several different discordance signals



Fontaine et al (Fig 1C, 2015)

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What is introgression?

- Two species can occasionally hybridize — mate and produce offspring.
- If hybrids repeatedly backcross into one parent, DNA from the other becomes permanently incorporated.

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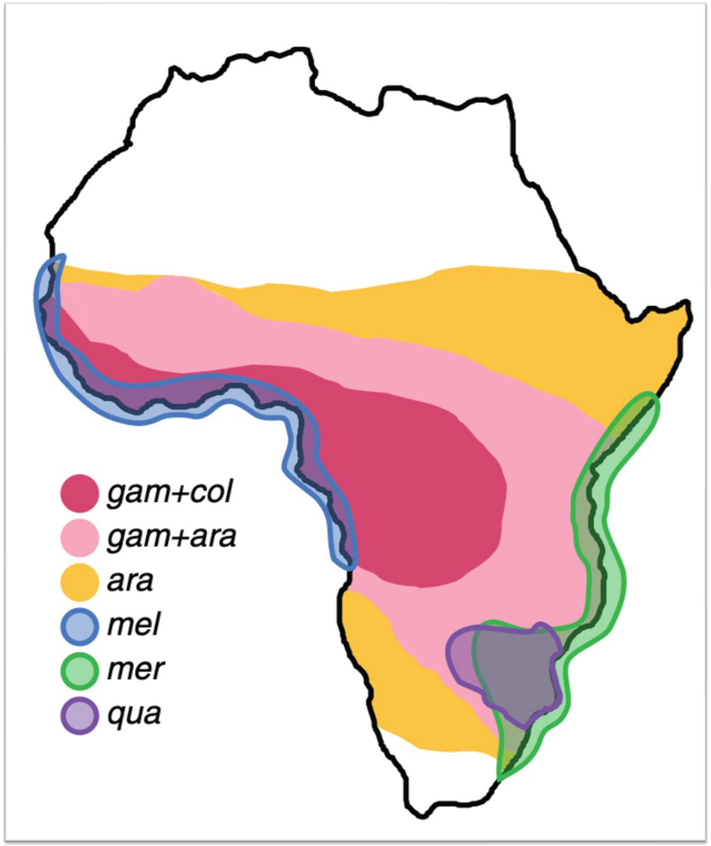
Six species, one complex

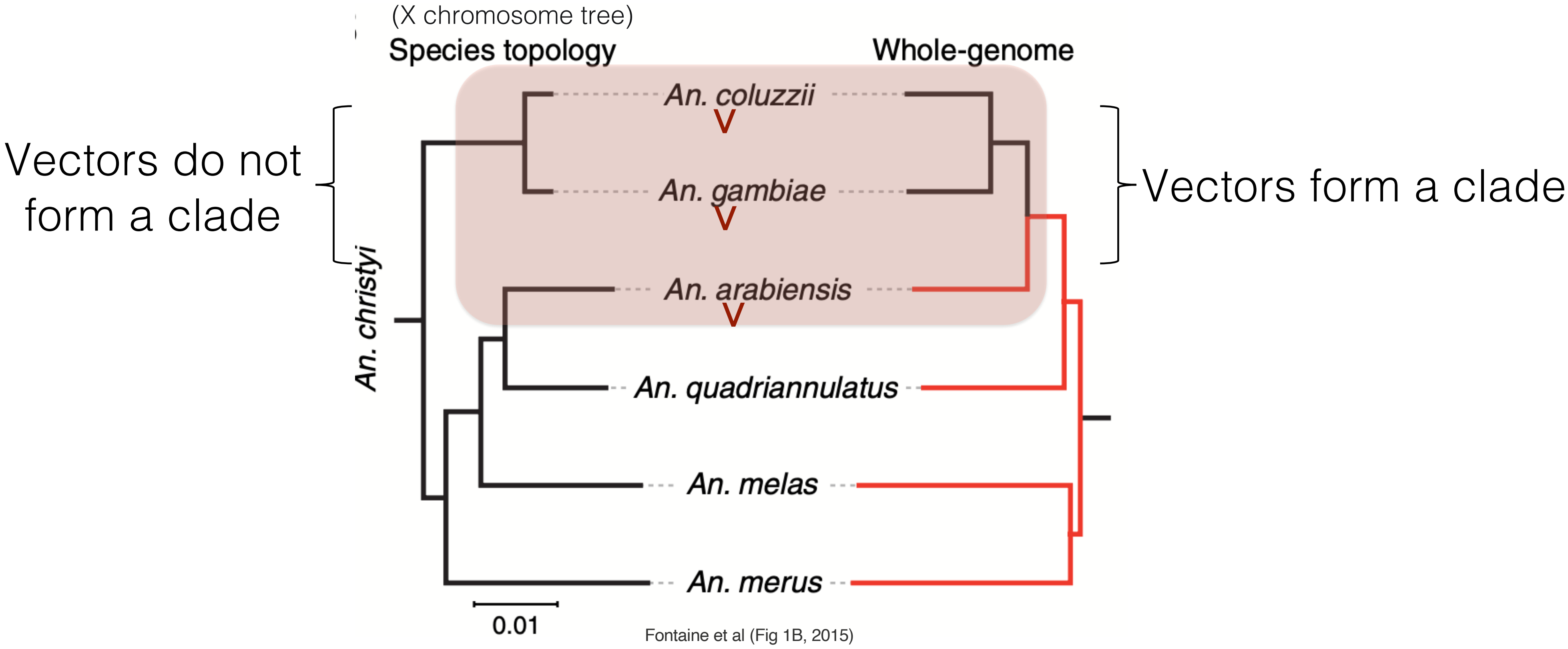
- *Anopheles gambiae* (vector)
- *An. coluzzii* (vector)
- *An. arabiensis* (vector)
- *An. quadriannulatus* (non-vector)
- *An. melas* (minor vector, brackish)
- *An. merus* (minor vector, brackish)

- *An. christyi* (outgroup)



Fontaine et al (Fig 1A, 2015)

The same dataset, two different trees



- Both trees 100% bootstrap support, which is “correct”?

Key findings (Fontaine et al, 2015)

- The X-chromosome tree is the species tree
  - higher divergence between sister pairs confirms the branching order.
- Autosomes are pervasively introgressed between
  - *An. arabiensis* and (*An. gambiae* + *An. coluzzii*).
- Vectorial capacity may have spread by introgression, not only de novo mutation.

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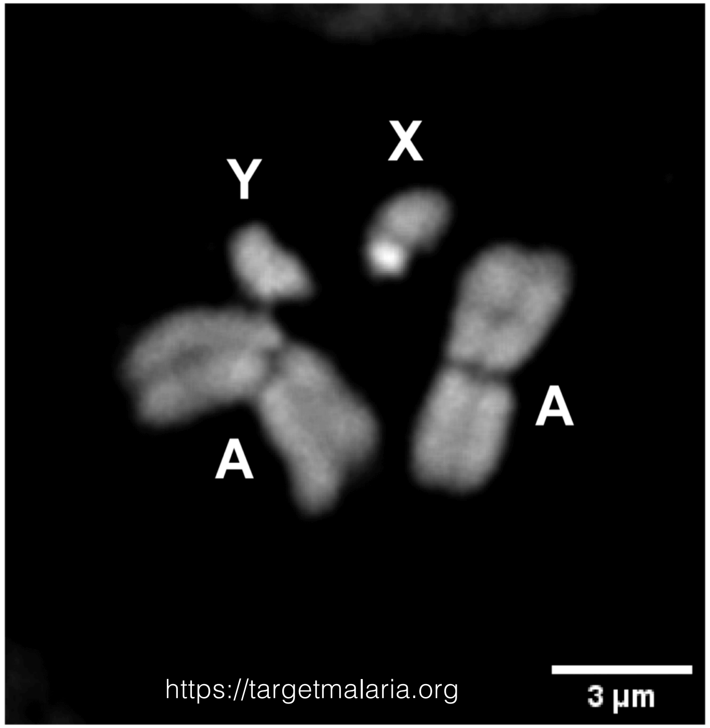
Tutorial data: 15 loci sampled to show the story



Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
Autosomal (2R, 3R)	5	(arabiensis, (gambiae, coluzzii)) - introgressed
2La inversion (2L)	2	2La-specific topologies
3La inversion (3L)	2	(merus, quadriannulatus) - unexpected introgression

Names such as 2La or 3La refer to specific chromosomal inversions

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Software for trees and networks

Each tool does one job; together they form the workflow.

Part I (trees)

IQ-TREE 3

per-locus ML

BEAST X

Bayesian inference

- BEAUti,
- Tracer,
- TreeAnnotator

ASTRAL

MSC species tree

Part II (networks)

SplitsTree

Consensus networks,

Neighbor-Net

PhyloSketch

interactive sketching

PhyloParallelograms

gene-trees-to-network

Throughout

- PhyloGuide

AI assistant

What we will do today

When the genome is pervasively introgressed, tree methods can be confidently wrong. Networks make the conflict visible.

- Part I: infer gene trees and a species tree from a multi-locus alignment — and see them disagree across the genome.
- Part II: use phylogenetic networks to visualize and interpret that disagreement.

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Quick setup verification

- ☐ Repository cloned:

```
git clone https://github.com/husonlab/trees-to-networks-tutorial.git
```

- ☐ `iqtree3 --version` works
- ☐ `astral -help` or `java -jar /path/to/astral.5.7.8.jar -h` works
- ☐ `beauti` and `beast` launch
- ☐ `SplitsTree` launches
- ☐ `tracer` launches
- ☐ `PhyloSketch` launches
- ☐ `PhyloParallelograms` launches
- ☐ `PhyloGuide` URL bookmarked

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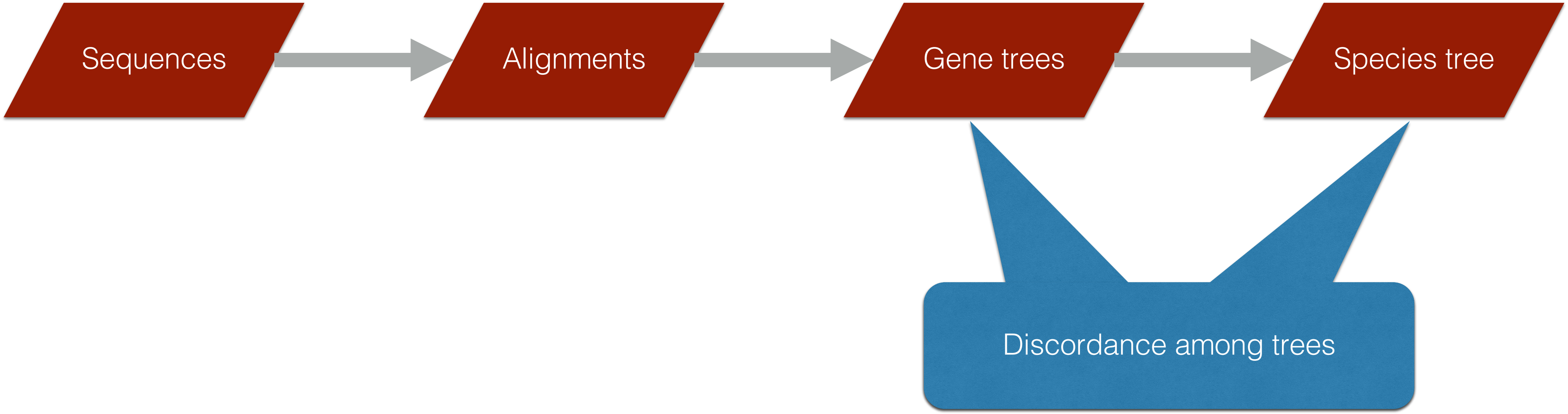
From sequences to gene trees to species trees



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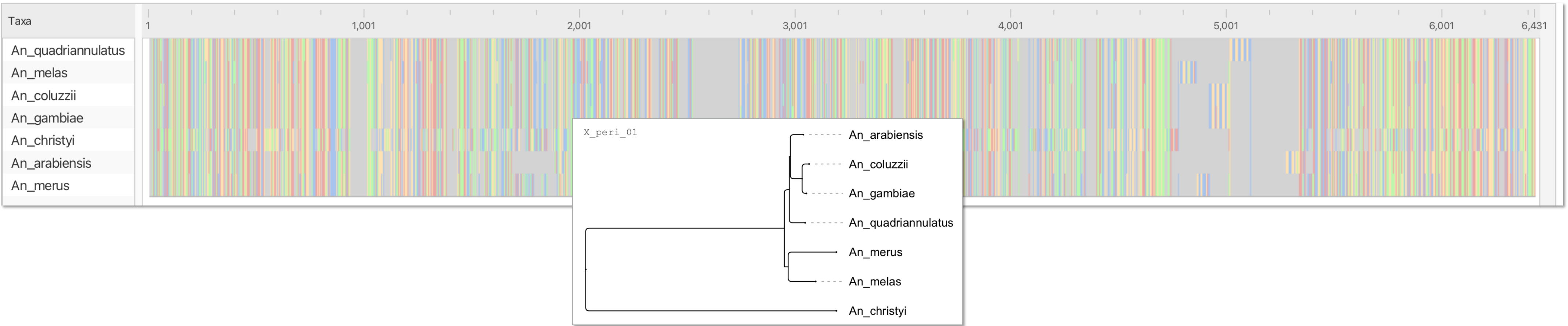
At a single locus

- Input: a multiple sequence alignment (MSA) — 7 species, ~5000 nt, every column informative.
- From the MSA we infer a gene tree: the history of this one stretch of DNA.



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Inferring a tree by maximum likelihood

- Probability of alignment A given tree T and model M:

$$P(A \mid T, M)$$

- Find the tree and model that maximize it — with IQ-TREE 3.
- Substitution models (JC, K2P, HKY, GTR, +G, +I) describe how nucleotides change.
- Bootstrap (UFBoot) assesses confidence in the tree.

Theory: Felsenstein (1981)
Implementation: IQ-TREE 3 (2026)
Enhancements: ModelFinder (2017), **UFBoot2 (2018)**

Inferring a *distribution* of trees by Bayesian inference

- Bayesian inference estimates the full posterior

$$P(T, M \mid A),$$

  not just one best tree.
- Tool: BEAST X (MCMC) — thousands of trees, each weighted by posterior probability.
- Captures uncertainty.

Theory: Yang & Rannala (1997)
Implementation: Baele et al. (2025)

Genome-scale data: many loci, many gene trees

- Phylogenomics: hundreds to thousands of loci, each with its own history.
- Run ML or Bayesian inference per locus → a set of gene trees.
- Our tutorial: 15 loci, 15 gene trees.

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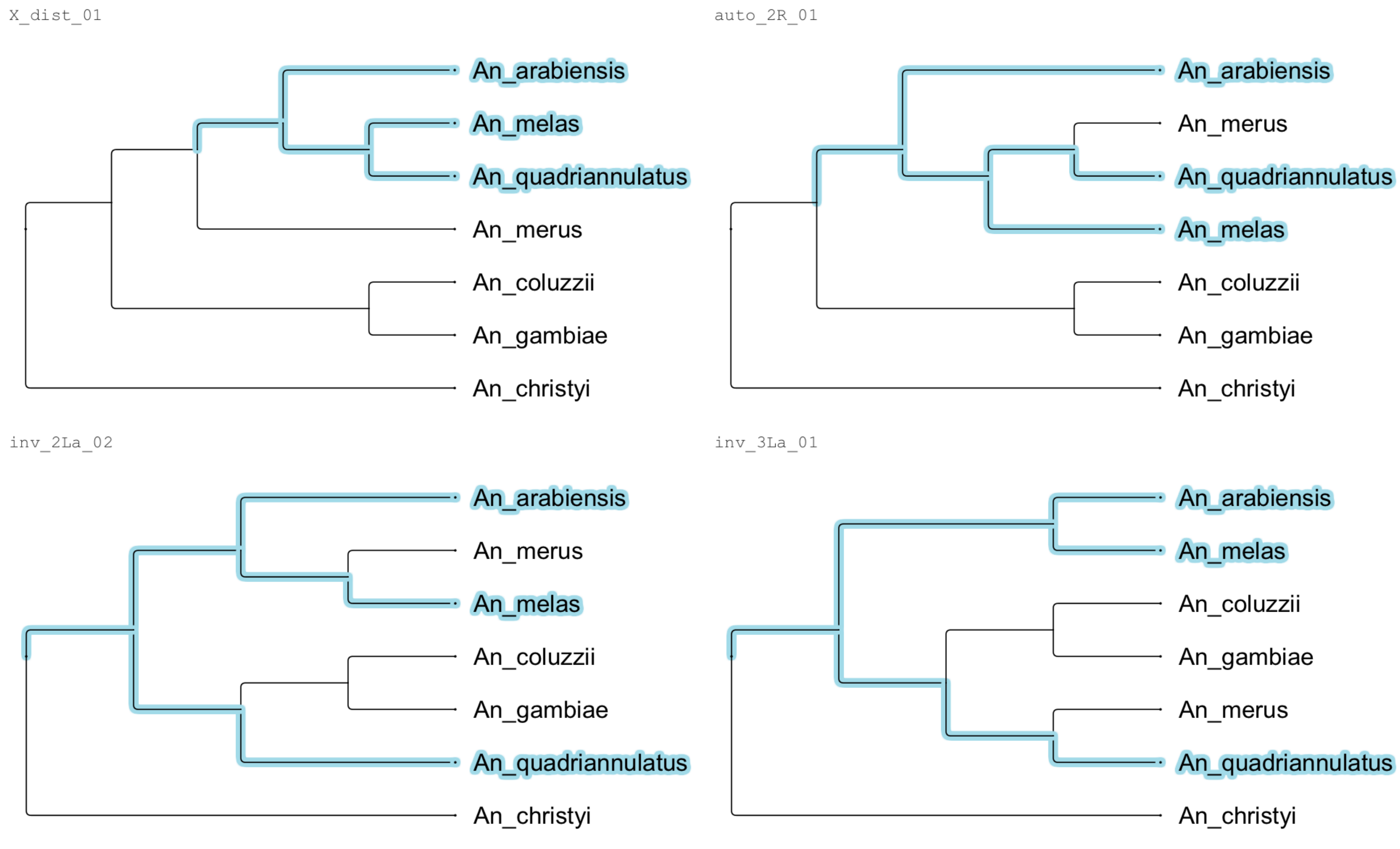
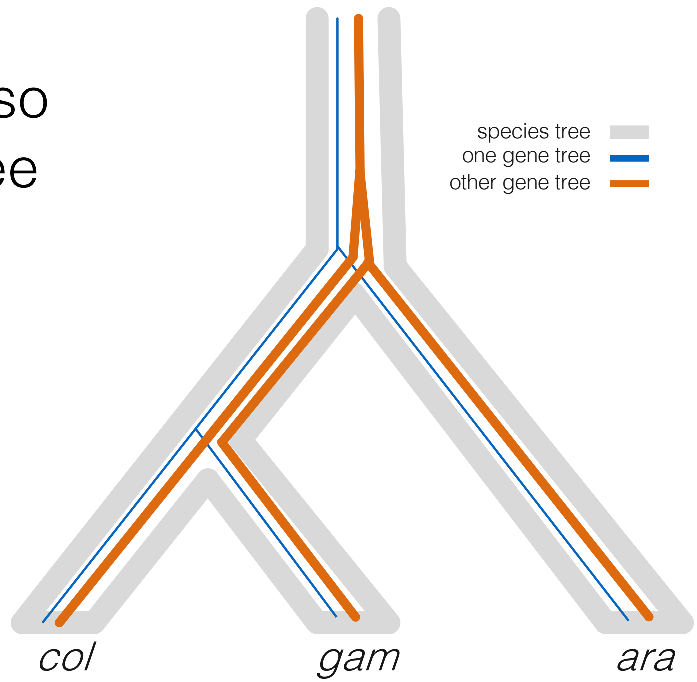
Why do gene trees disagree?



Cause #1 of discordance: incomplete lineage sorting (ILS)

- Ancestral polymorphism can persist across several species
- Different alleles sort into different lineages, so the gene tree can differ from the species tree
- A stochastic process — no gene flow



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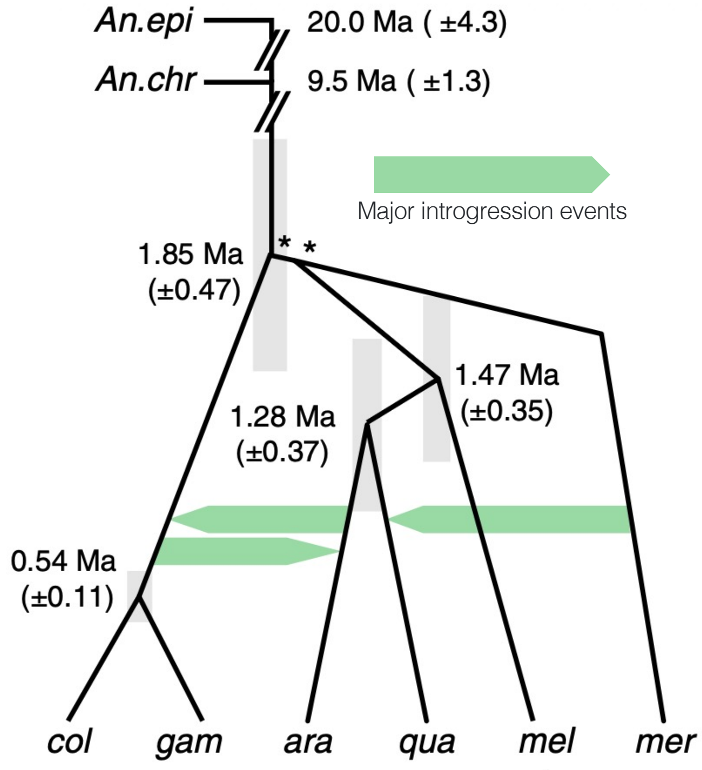
Cause #2 of discordance: introgression

- After speciation, gene flow (hybridization or HGT) can reintroduce alleles
- Introgressed alleles resemble the donor, not the closest relative
- Often non-uniform across the genome
- A biological process



Fontaine et al (Fig 1C, 2015)

Telling ILS and introgression apart

- Both ILS and introgression make gene trees disagree with the species tree.
- ILS: unsorted alleles are ancestral → coalesce *deep* in the species tree.
- Introgression: alleles came recently from a sister species → coalesce *shallow*.
- This is how Fontaine et al. (2015) showed *Anopheles* autosomes are introgressed, not just ILS.

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Now: let's try to see this in our data

- 15 loci across the X chromosome, autosomes, and inversions.
- Run IQ-TREE on each → 15 gene trees.
- We expect discordance — the question is *where?*

Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
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PhyloGuide Q1

PhyloGuide
PHYLOGENETICS, GROUNDED

Q: We have multi-locus alignments from a recently diverged species complex. How should we infer a species phylogeny?

A: Infer gene trees for each locus, then estimate a species tree with ASTRAL instead of relying only on concatenation. This accounts for gene-tree discordance caused by ILS.

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Hands-on: per-locus ML inference with IQ-TREE 3

- Time budget: ~20 minutes
- Goal: 15 ML gene trees, one per locus — see how they vary across the genome.
- `cd trees-to-networks-tutorial`
- Run: `iqtree3 -S data/alignments/` ... (see next slide)

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The IQ-TREE 3 command

```
iqtree3 -S data/alignments -B 1000 --prefix tutorial_loci -T AUTO -st DNA
```

Flag	What it does
`-S data/alignments`	Per-locus mode: one tree per FASTA in this directory
`-B 1000`	Ultrafast bootstrap with 1000 replicates
`--prefix tutorial_loci`	Output filename prefix
`-T AUTO`	Auto-pick number of threads
`-st DNA`	Force DNA mode (auto-detection fails on some loci)

Run the IQ-TREE 3 command

- Run the command.
- Internally: ModelFinder picks a model per locus, then ML tree search, then UFBoot support.
- Runtime ~5 s. Output: `tutorial_loci.treefile`

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Run the IQ-TREE 3 command

- You'll see one block of output per locus, ending like this:

```
Subset   Type    Seqs    Sites   Infor   Invar   Model   Name
1    DNA     7   6208    144     5068      X_dist_01.fasta
2    DNA     7   7530    234     5899      X_dist_02.fasta
...

15     DNA  7     5454 105   4455      inv_3La_02.fasta
```

- What's normal:
  - All 15 loci have 7 sequences; most pick HKY+F+G4
  - *An. christyi* (the outgroup) failing composition chi2 tests in several loci is not a problem

Nexus file for SplitsTree and PhyloParallelograms

- IQ-TREE wrote 15 Newick trees to `tutorial_loci.treefile`, one per line in alphabetical order of FASTA filenames
- For SplitsTree and PhyloParallelograms, we want a NEXUS file with each tree labeled by its locus ID and correctly rooted by outgroup An_christyi
- Run our helper:

```
python tools/make_nexus_treeset.py \
    --alignments-dir data/alignments/ \
    --treefile tutorial_loci.treefile \
    --out tutorial_loci.nex\
    --outgroup An_christyi
```

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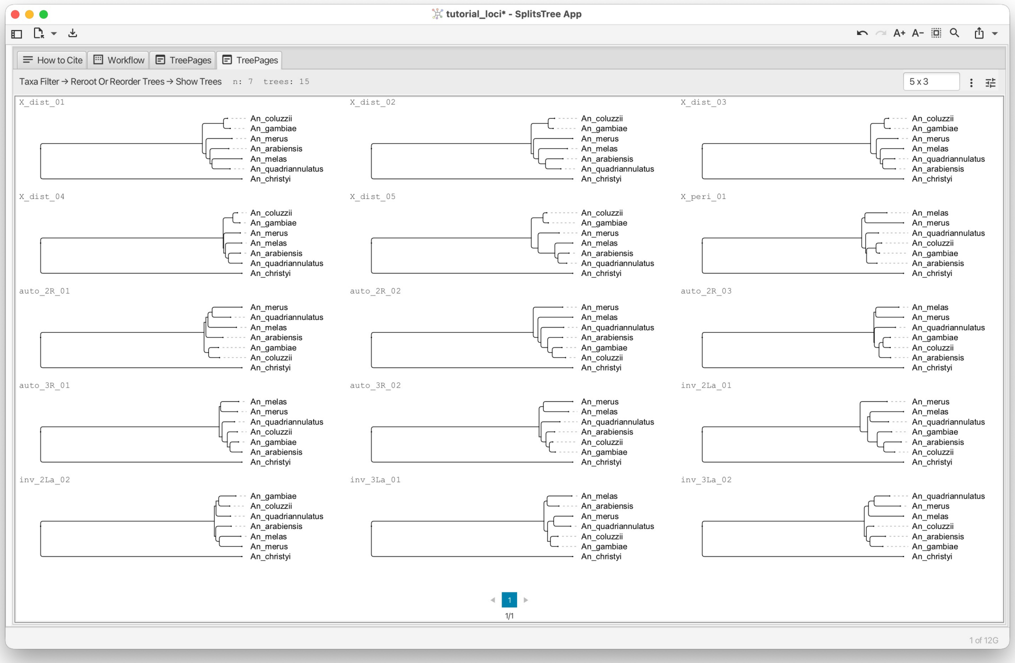
Open the gene trees in SplitsTree and look around

- Open `tutorial_loci.nex` in SplitsTree
  - File -> Open -> select the .nex file
- What you should see: 15 named trees in the tree set



Discussion items

- Do all loci show the same topology? Why not?
- Can you tell apart X-distal from autosomal loci just by topology?
- Where in the genome would you expect to see "species-tree" signal vs "introgressed" signal, based on what we know about this dataset?

PhyloGuide Q2

PhyloGuide
PHYLOGENETICS, GROUNDED

Q: We ran IQ-TREE on 15 alignments and the gene trees disagree across chromosomes - X chromosome shows one topology, autosomes another. What does this mean?

A: Different genomic regions can have different evolutionary histories due to ILS, introgression, selection, or gene-tree error. It does not mean one topology is necessarily correct.

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Hands-on: Bayesian inference + visualizing the posterior

- Time budget: 30 minutes
- Goal: build a posterior tree distribution for one locus, then visualize within-locus uncertainty as a network.
- Tools: BEAST X, BEAUti, Tracer, TreeAnnotator, SplitsTree
- Locus: X_dist_04 (a clean X-distal alignment)

BEAST X
Bayesian Evolutionary Analysis Sampling Trees

From “best tree” to “distribution of trees”

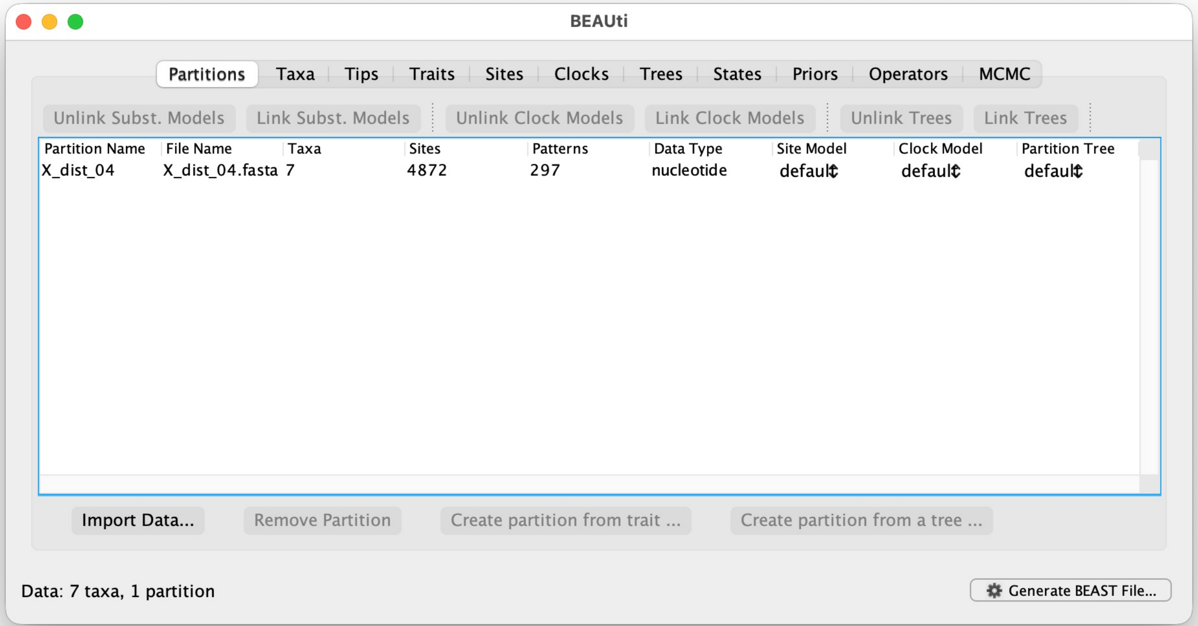
- What ML gave us (IQ-TREE):
  - One tree per locus, with bootstrap support
  - Discordance ACROSS loci
- What Bayesian gives us (BEAST X):
  - A posterior distribution of trees per locus — uncertainty WITHIN a locus
  - Summarize as one tree, or visualize the full distribution

We’ll run BEAST X on a single locus: X_dist_04

- Locus: chrX:9,845,592–9,849,716 (4,872 bp), X-distal Xag region
- Expected topology: (arabiensis, quadriannulatus) sister, christyi outgroup; model HKY+F+G4.
- Why this locus: clean signal, fast convergence, the true-species-tree topology — one locus keeps the demo short.

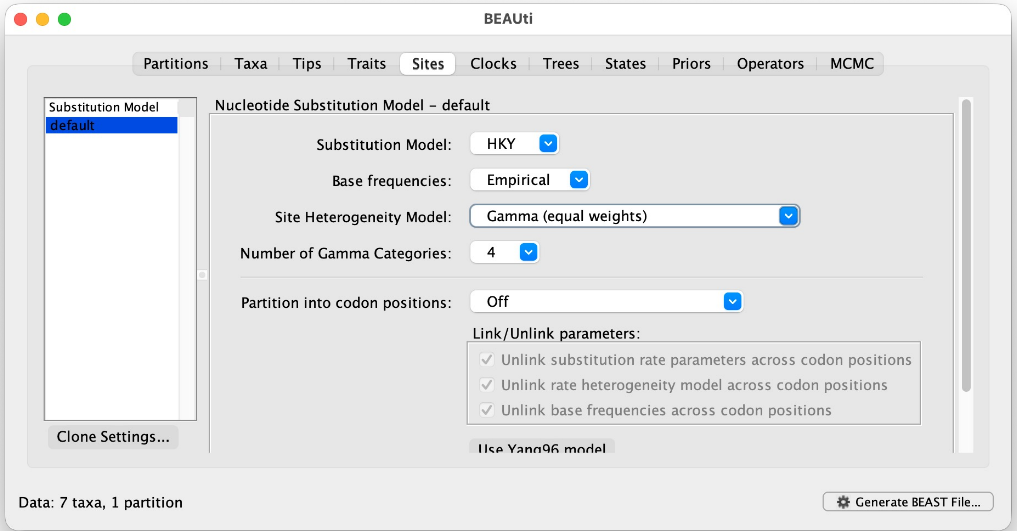
BEAUti: import data and set substitution model

1. **Launch BEAUti** (from BEAST X installation)
2. **File -> Import Data ->** `data/alignments/X_dist_04.fasta`



BEAUti: import data and set substitution model

3. **Sites tab**:
   - Substitution Model: **HKY**
   - Base frequencies: **Empirical**
   - Site Heterogeneity Model: **Gamma**
   - Number of Gamma Categories: **4**
   - Partition into codon positions: **off**



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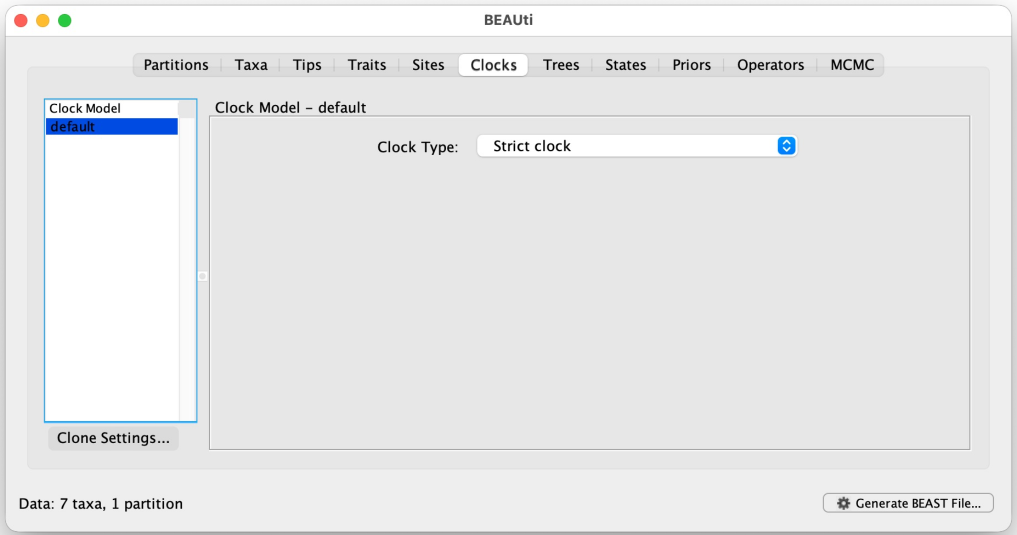
BEAUti: clock model and tree prior

4. Clocks tab:
   - Clock Type: **Strict clock**



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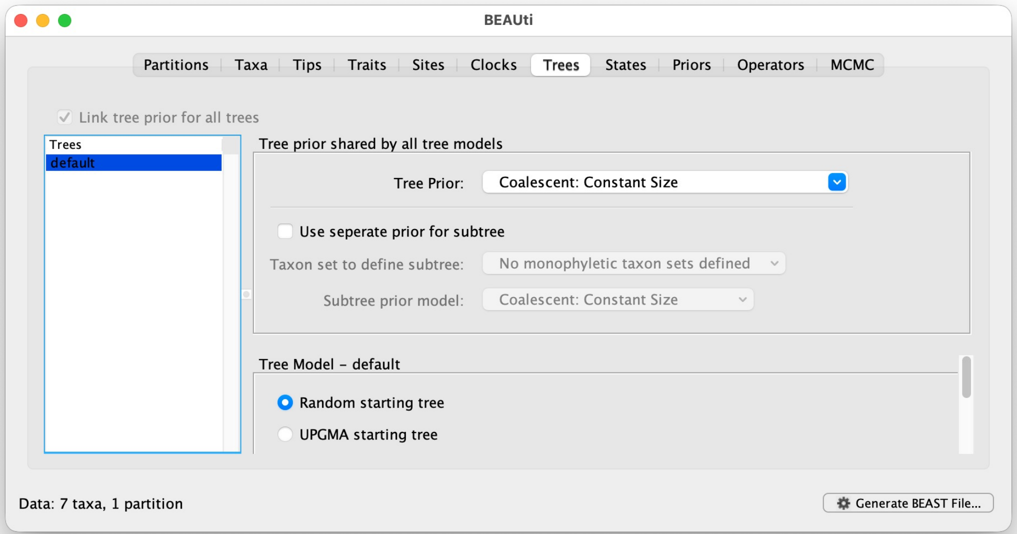
BEAUti: clock model and tree prior

5. Trees tab:
   - Tree Prior: **Coalescent: Constant Size**
   - This is appropriate for the within-genus timescale



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BEAUti: clock model and tree prior

6. **States tab**: nothing to change
7. **Priors tab**: leave at defaults
8. **Operators tab**: leave at defaults

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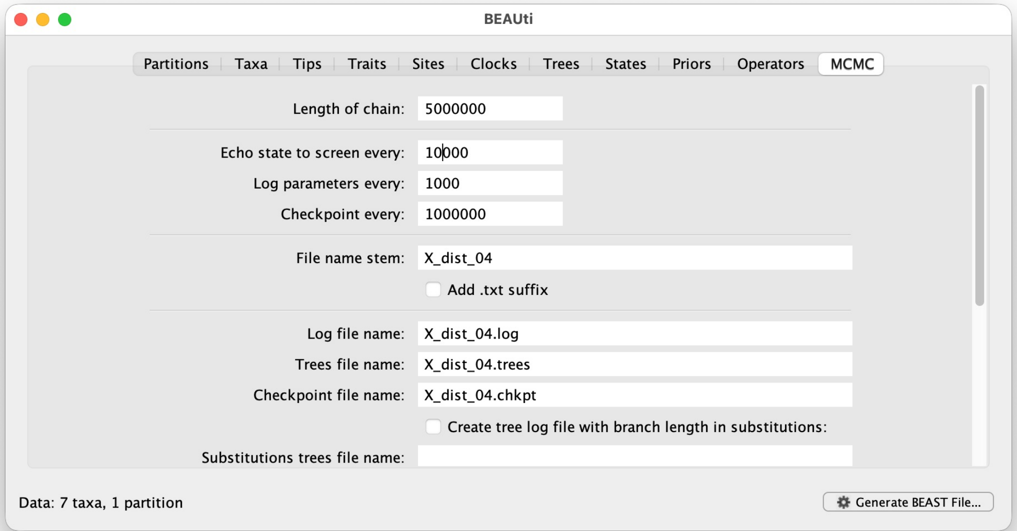
MCMC settings, generate XML, and start BEAST

9. MCMC tab:
   - Length of chain: **5,000,000** (short for live demo; production runs would use 20M)
   - Echo state to screen every: 10,000
   - Log parameters every: **1000** (gives 5,000 samples)
   - File name stem: **X_dist_04**



MCMC settings, generate XML, and start BEAST

10. File -> Generate BEAST File -> save as `X_dist_04.xml`
11. Run BEAST X immediately in a terminal:

```
beast -beagle -overwrite X_dist_04.xml
```

- Expected runtime: about 2-3 minutes on Apple Silicon with BEAGLE
- Let it run in the background. We will come back to it later.

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Inspect a precomputed run in Tracer (while yours runs)

- Your live run is writing `X_dist_04.log` and `X_dist_04.trees`, but is not done yet
- To see good convergence now, open the precomputed run:

```
Tracer

File -> Import Trace File

-> precomputed/beast/X_dist_04.log
```

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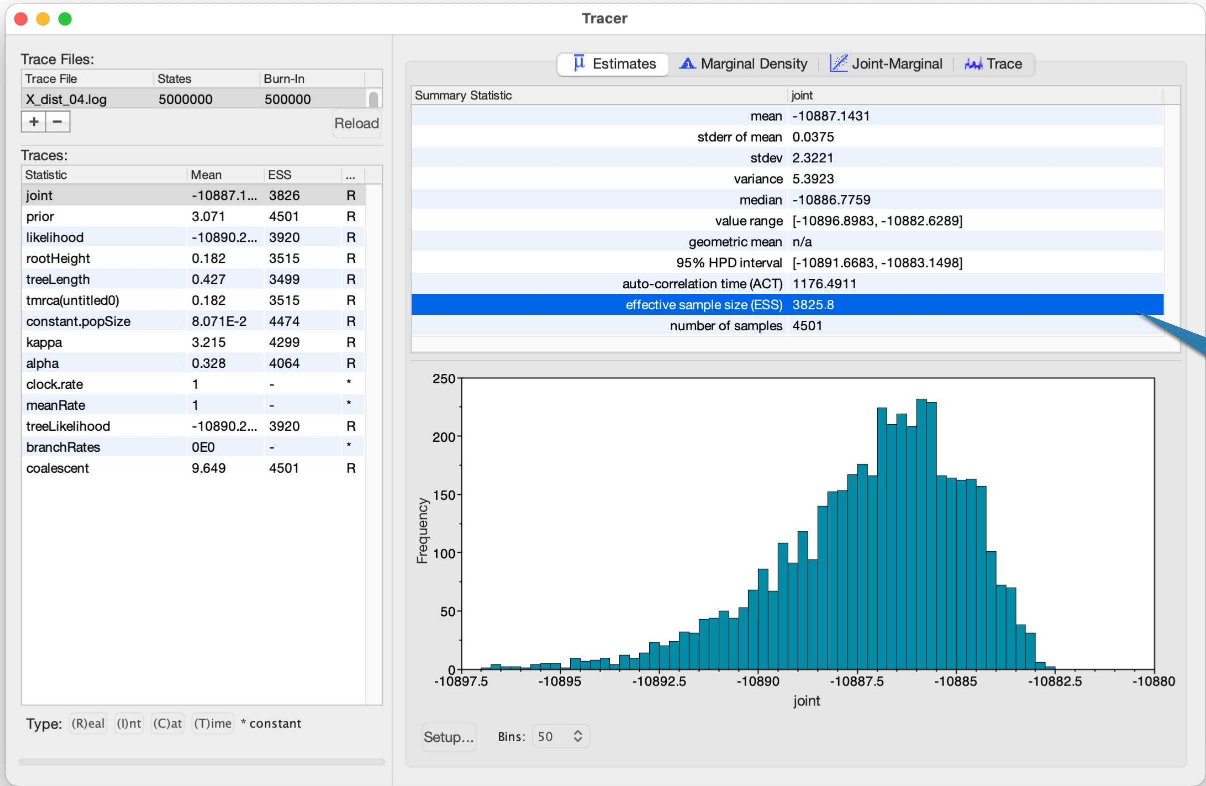

Effective sample size is good (nearly 4000)

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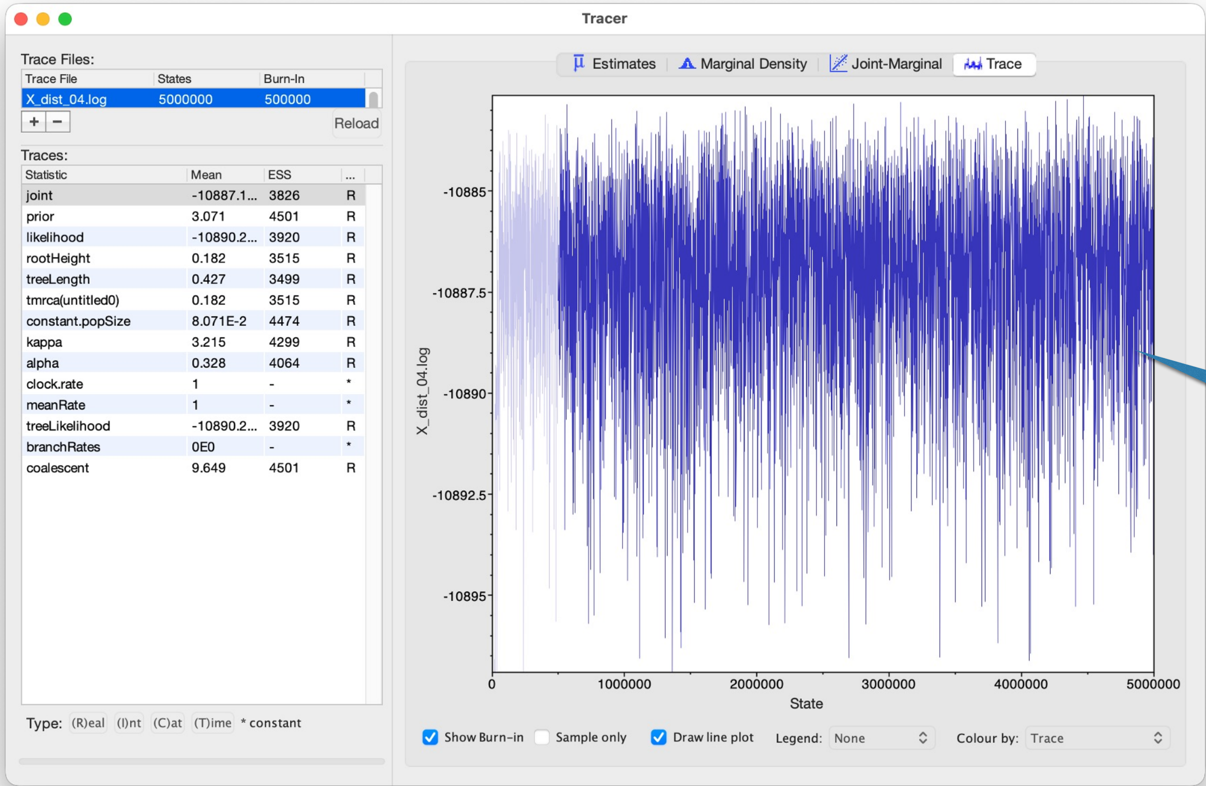
Inspect a precomputed run in Tracer (while yours runs)



Sampling looks good
(“hairy caterpillar”)

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Summarize the posterior with TreeAnnotator

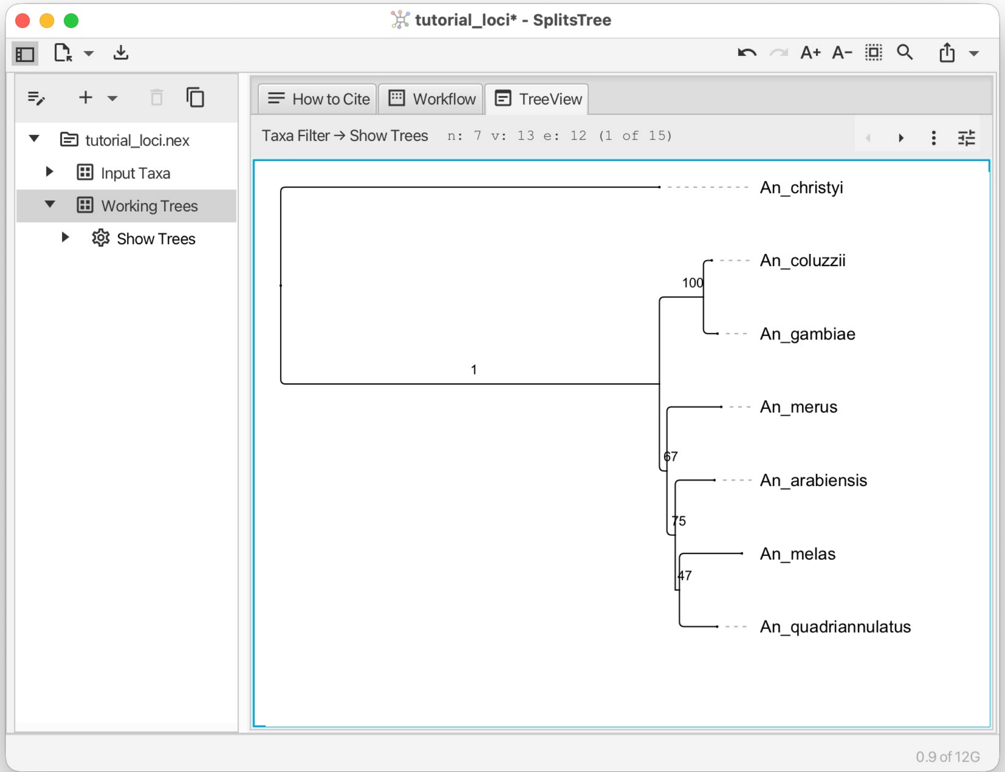
- We have a distribution of trees but often want one summary tree
- TreeAnnotator computes the Maximum Clade Credibility (MCC) **tree**: the tree from the posterior whose clades collectively have the highest product of posterior support

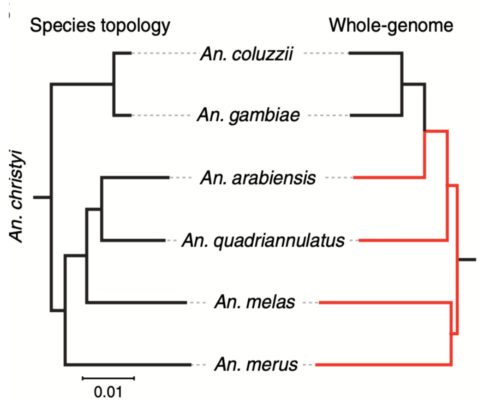
```
treeannotator -burnin 500 -heights median \
    precomputed/beast/X_dist_04.trees X_dist_04.MCC.tre
```

`-burnin 500`	Discard first 500 trees
`-heights median`	use median node heights for branch lengths
`precomputed/beast/X_dist_04.trees`	Input trees
`X_dist_04.MCC.tre`	one tree with posterior probabilities annotated on each internal node

Summarize the posterior with TreeAnnotator

- Open in SplitsTree:





Fontaine et al (Fig 1B, 2015)

- MCC keeps (arabiensis, quadriannulatus) and (coluzzii, gambiae) — the species signal — but the deep backbone (melas, merus) is poorly resolved.

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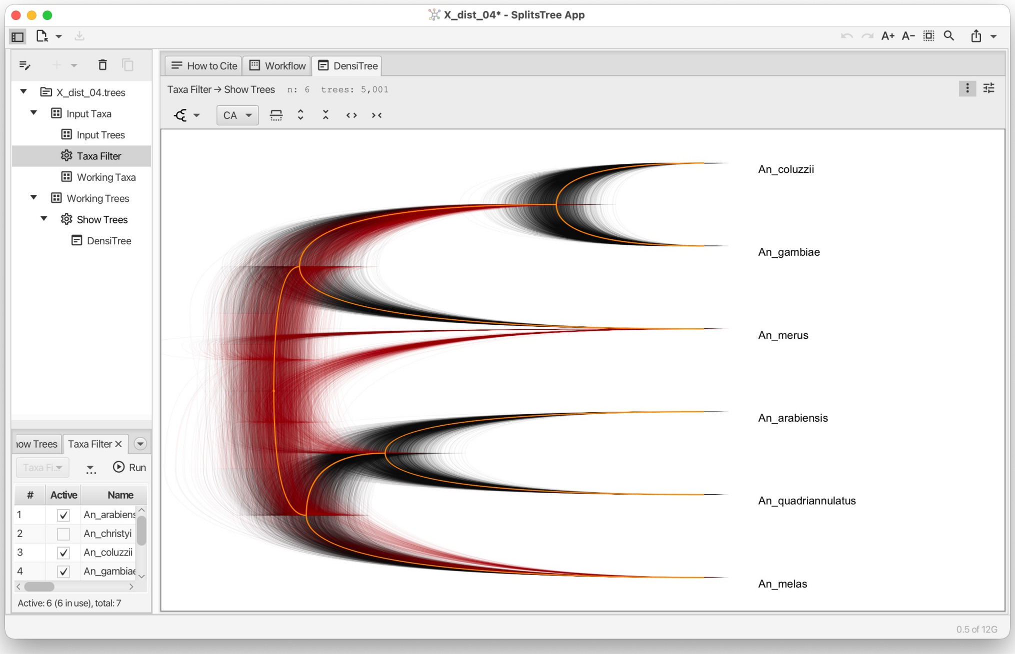
Visualizing the full posterior: DensiTree in SplitsTree

By the time you see this slide, your live BEAST run should be done.

- Open `X_dist_04.trees` in SplitsTree (the full posterior, 5000 trees)
- Tree -> Show DensiTree to overlay all trees



Your first network: consensus outline from the posterior

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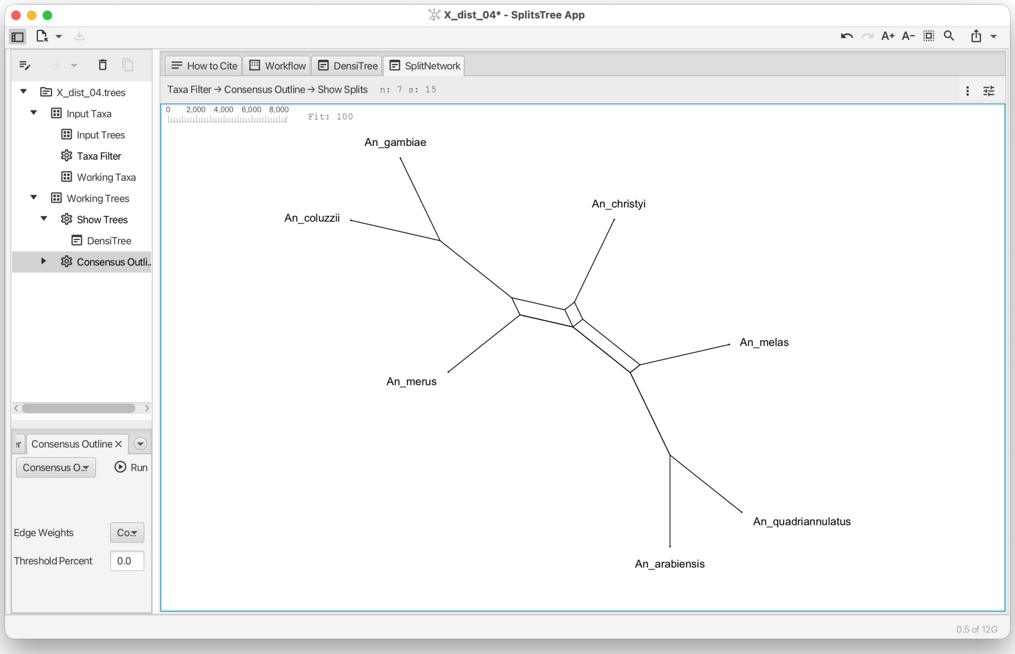
- In SplitsTree, with the same posterior tree set loaded:
  - Network menu > Consensus Outline

➢ Network with splits weighted by posterior support (counts)



Here:
Boxes= posterior uncertainty,
not hybridization.

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Hands-on: coalescent species tree with ASTRAL

- ASTRAL: for a set of gene trees, returns a single species tree that maximizes quartet agreement under the multispecies coalescent (it models ILS but NOT introgression)
- **Input**: `tutorial_loci.treefile` (output of IQ-TREE)
- **Command**:

```
java -jar astral.5.7.8.jar \
   -i tutorial_loci.treefile \
   -o tutorial_loci.ASTRAL_species_tree.tre
```

➢ Species tree with quartet support values at each internal node

Maddison (1997), ASTRAL-III (Zhang et al., 2018)

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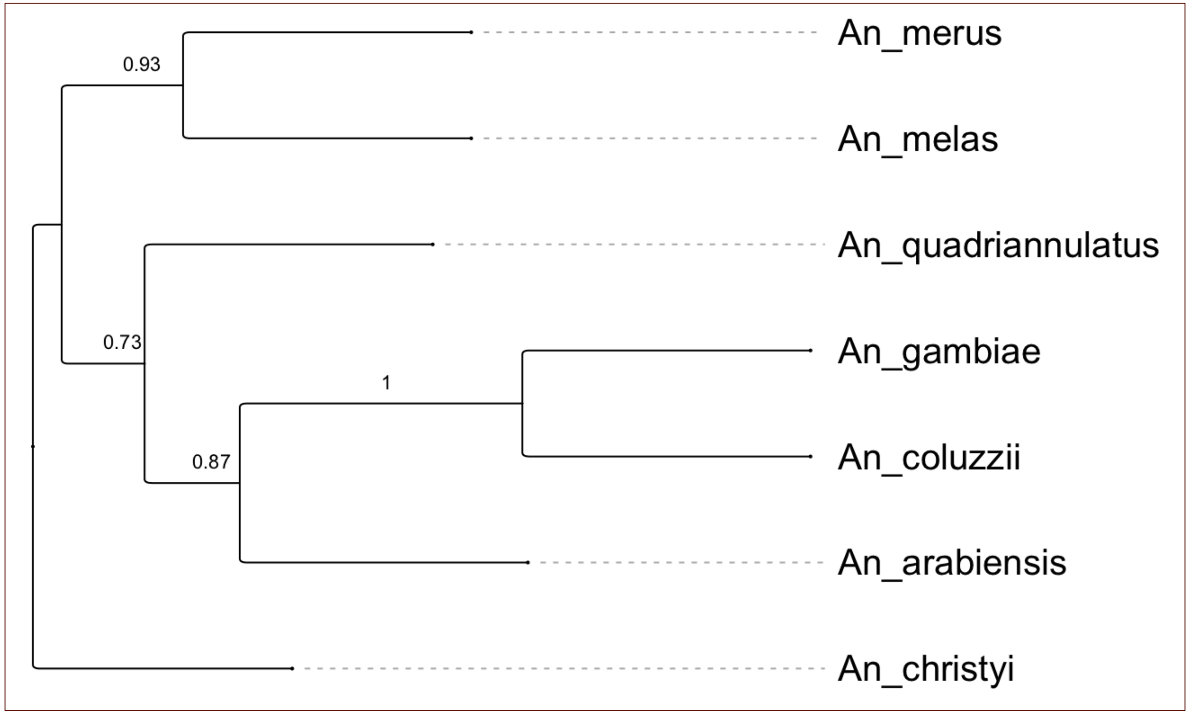
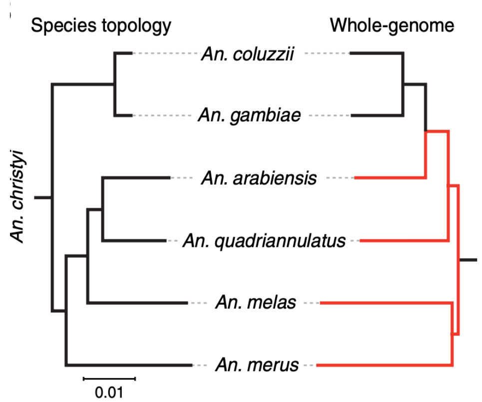
What does ASTRAL tell us?

Open the ASTRAL output in SplitsTree and reroot using An_christyi as outgroup:





Fontaine et al (Fig 1B, 2015)

This is the introgressed phylogeny: *An. arabiensis* sister of (*An. gambiae* and *An. coluzzii*)

ASTRAL Is misled by autosomal introgression

- The X-distal gene trees give the (arabiensis, quadriannulatus) topology — the true species branching order (Fontaine 2015).
- The autosomal trees and ASTRAL give the introgression-driven topology instead.
- ASTRAL gave a confident answer that is probably biologically wrong.
- It is not broken — MSC models ILS only, and cannot represent the introgression in our data.

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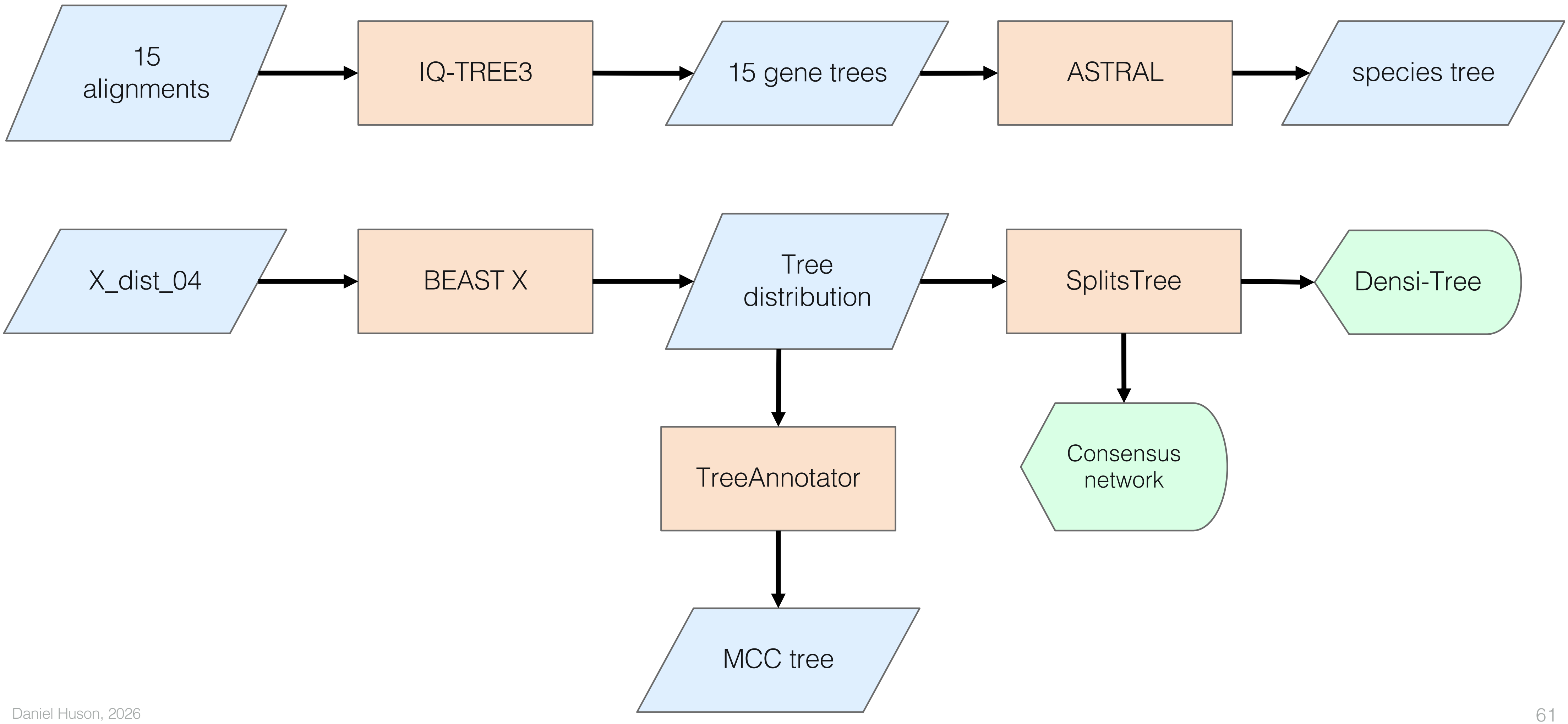
Part I recap: from sequences to trees



PhyloGuide Q3

PhyloGuide

PHYLOGENETICS, GROUNDED

Q: When is a tree inadequate? When do we need a network?

A: A tree is inadequate when evolution is not strictly branching, such as with hybridization, introgression, HGT, or strong conflicting signal. Networks capture mixed ancestry and conflicting histories.

After the break: Making conflict explicit

- **Part II preview**:
  - SplitsTree Neighbor-Net: networks of conflicting splits in the alignment data
  - PhyloSketch: build a hypothesis network interactively
  - PhyloParallelograms: networks inferred from the gene trees - and SEE the introgression
- **Break**: 20 minutes, then 11:10 reconvene

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COFFEE BREAK

20 minutes

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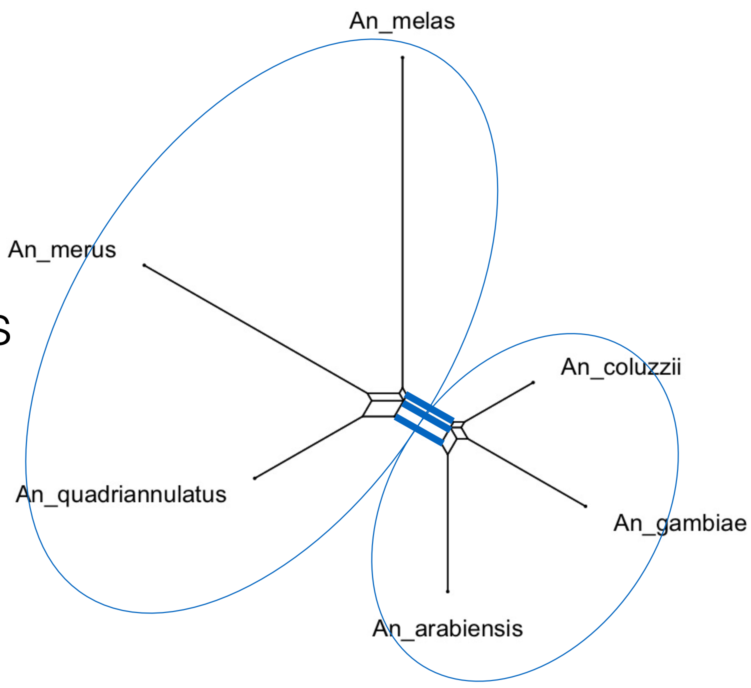
Part II: Networks and cross-locus reticulation

- Where we left off:
  - 15 gene trees disagree; ASTRAL's confident "species" tree is the introgression-driven one, not the true one.
  - The data carry a non-tree-like signal.
- What we will do next:
  - Three network views, each with a different tool — to see the introgression, not just infer it.

Split networks

- A split partitions the taxa into two sets; each tree edge is a split.
- A tree's splits are all compatible.
- A split network can show incompatible splits — any two produce a box.



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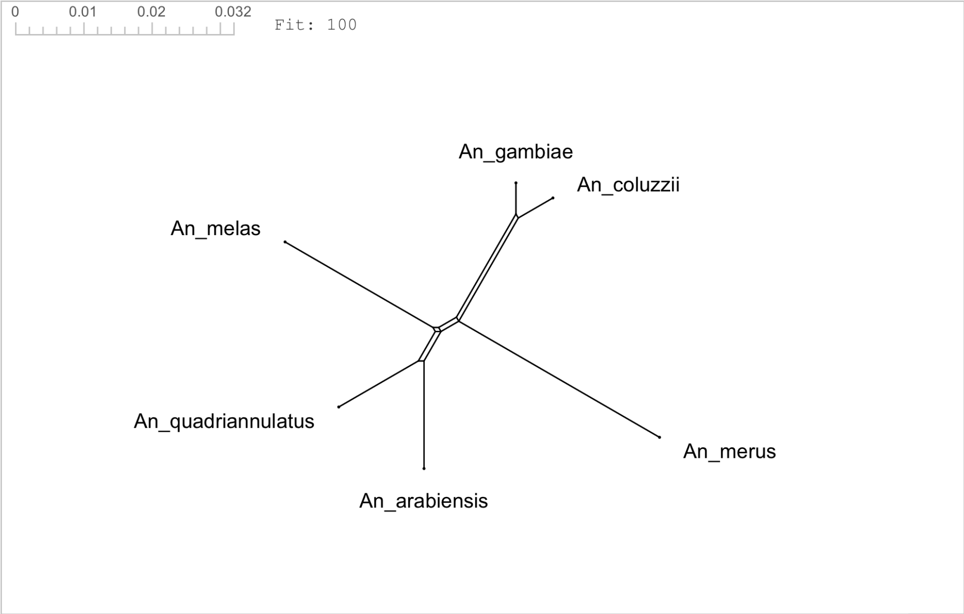
Building a split network from distances

- Input: Pairwise distances
- Neighbor-Net extends Neighbor-Joining, retaining conflicting evidence as boxes
- Output: a split network approximating the distances

'An_arabiensis'	0	0.04565785	0.02900961	0.04527742	0.04490629	0.05575238
'An_melas'	0.04565785	0	0.04266467	0.05110034	0.04913133	0.06128606
'An_quadriannulatus'	0.02900961	0.04266467	0	0.04385965	0.04351886	0.0543784
'An_coluzzii'	0.04527742	0.05110034	0.04385965	0	0.01068376	0.05556591
'An_gambiae'	0.04490629	0.04913133	0.04351886	0.01068376	0	0.05447761
'An_merus'	0.05575238	0.06128606	0.0543784	0.05556591	0.05447761	0



Bryant & Moulton (2004)

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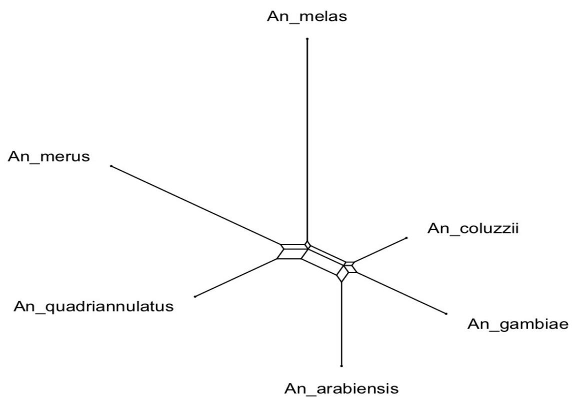
Two roles for networks: implicit and explicit

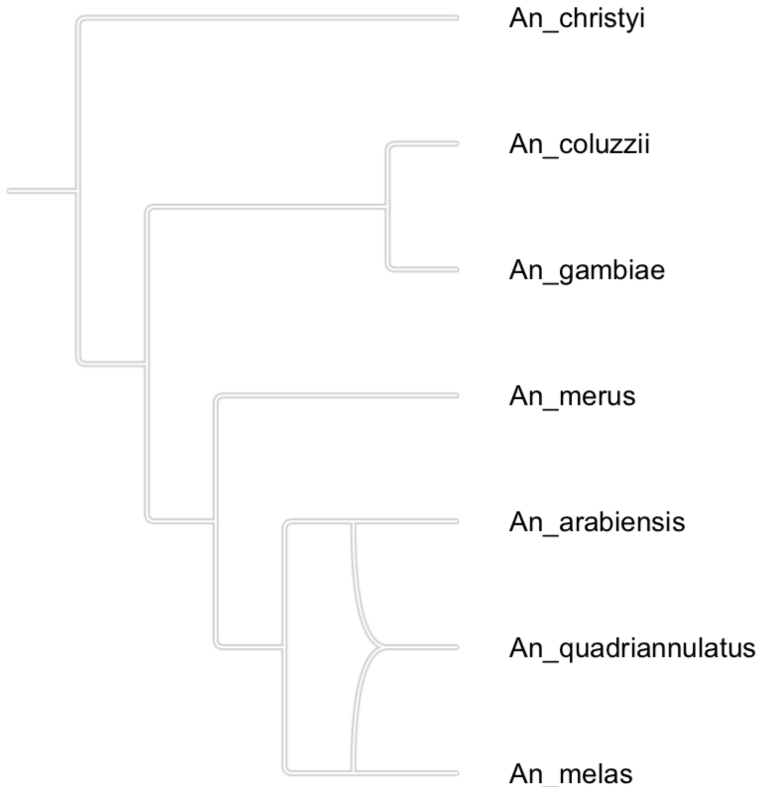
1. **Implicit networks** – data display networks

- Show conflicting signal without claiming specific events — “here is conflict”, not its cause.
- Example: split networks — for exploration and detecting non-tree-like signal.



Rooted phylogenetic networks

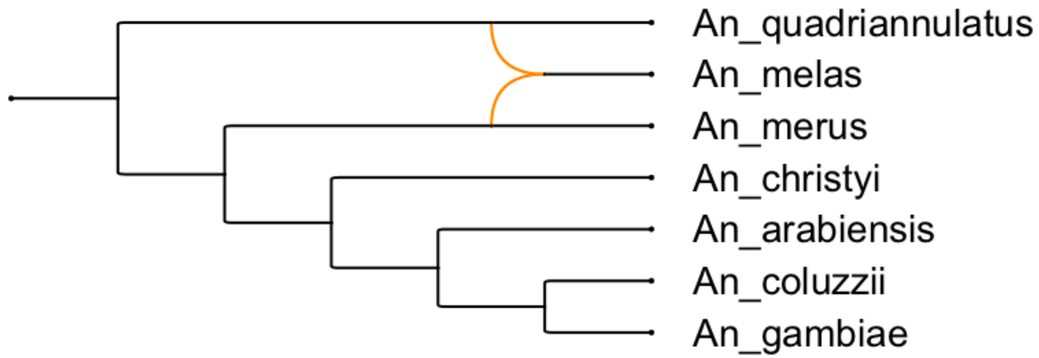
- A rooted tree: one root, leaves are taxa, internal nodes are inferred ancestors.
- A network generalizes this — internal nodes can have more than one parent (reticulation nodes).
- It captures both vertical inheritance and events where lineages combine or exchange DNA.



Two roles for networks: implicit and explicit

2. **Explicit networks** – explicit evolutionary scenario

- Represent specific events — hybridization, introgression, HGT — as reticulation nodes.
- “These two parent lineages produced this hybrid.”
- Example: PhyloFusion output.



Explicit reticulation: where two lineages meet

- A reticulation node = genetic material from two parent lineages combining into one descendant.
- May correspond to:
  - Hybridization, introgression, horizontal gene transfer, or recombination
- A network can have several reticulation nodes — one per event.

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You have already computed a network

- In Part I, the consensus network from the BEAST X posterior was an implicit network:
  - Splits weighted by posterior probability; boxes where it is uncertain (within-locus).
- Today we compute several more, viewing our 15 gene trees from different angles.

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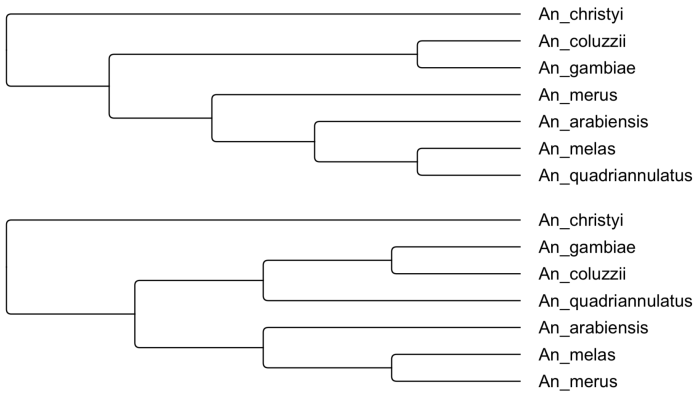
From gene trees to a reticulate network

- *PhyloFusion* takes rooted trees and computes an explicit network.
- It displays all input trees with as few reticulations as possible.
- Each reticulation is a hypothesized event explaining the disagreement.
- PhyloParallelograms: choose which trees to include and display.



Zhang, Cetinkaya, Huson (2026)

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Our three network views today

- 1. SplitsTree Neighbor-Net on alignments (20 min) — implicit split network; see raw conflict in the data.
- 2. PhyloSketch interactive sketching (20 min) — hand-draw a hypothesis network from your intuition.
- 3. PhyloParallelograms on the gene trees (30 min) — explicit reticulation network (PhyloFusion); see the introgression.

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Hands-on: Neighbor-Net on the 15 alignments

- Time budget: 20 minutes
- Goal: see conflicting signal directly in alignment data, before any gene tree (SplitsTree).
- Plan: run Neighbor-Net on one alignment together, then each load 2–3 more from other categories and compare.

Neighbor-Net on our alignments

- We use Neighbor-Net (SplitsTree) to compute split networks for our 15 alignments
- Question: do they all look the same? Where are the boxes?

Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
Autosomal (2R, 3R)	5	(arabiensis, (gambiae, coluzzii)) - introgressed
2La inversion (2L)	2	2La-specific topologies
3La inversion (3L)	2	(merus, quadriannulatus) - unexpected introgression

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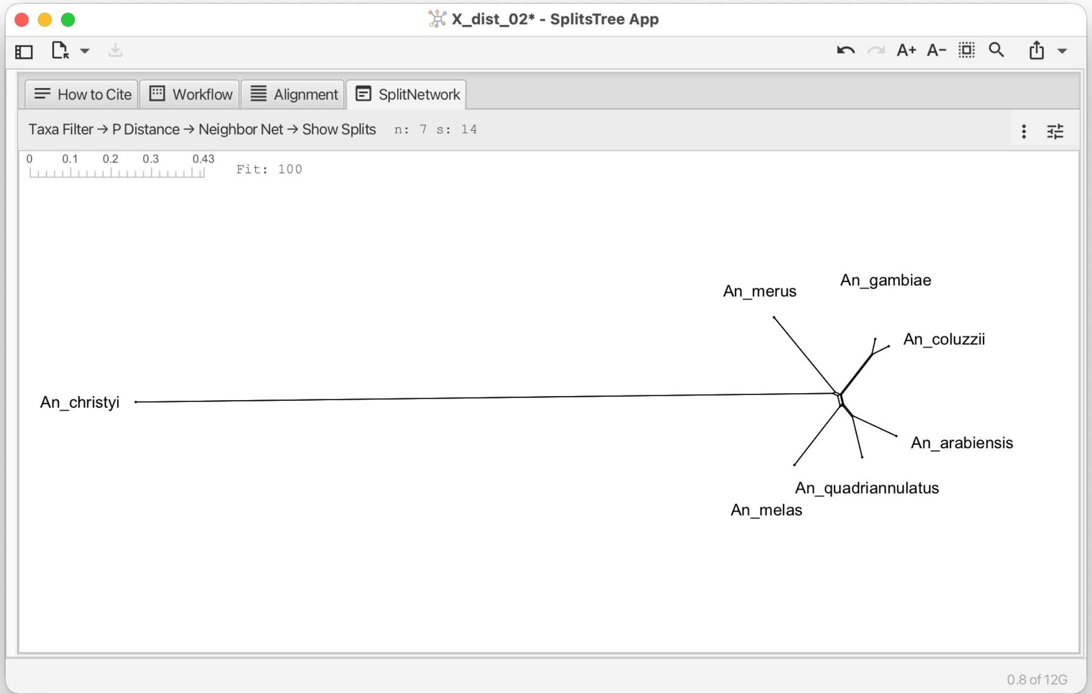
Let’s do one together: X_dist_02

- Open file `data/alignments/X_dist_02.fasta` in SplitsTree
- This will run p-distances and Neighbor-Net



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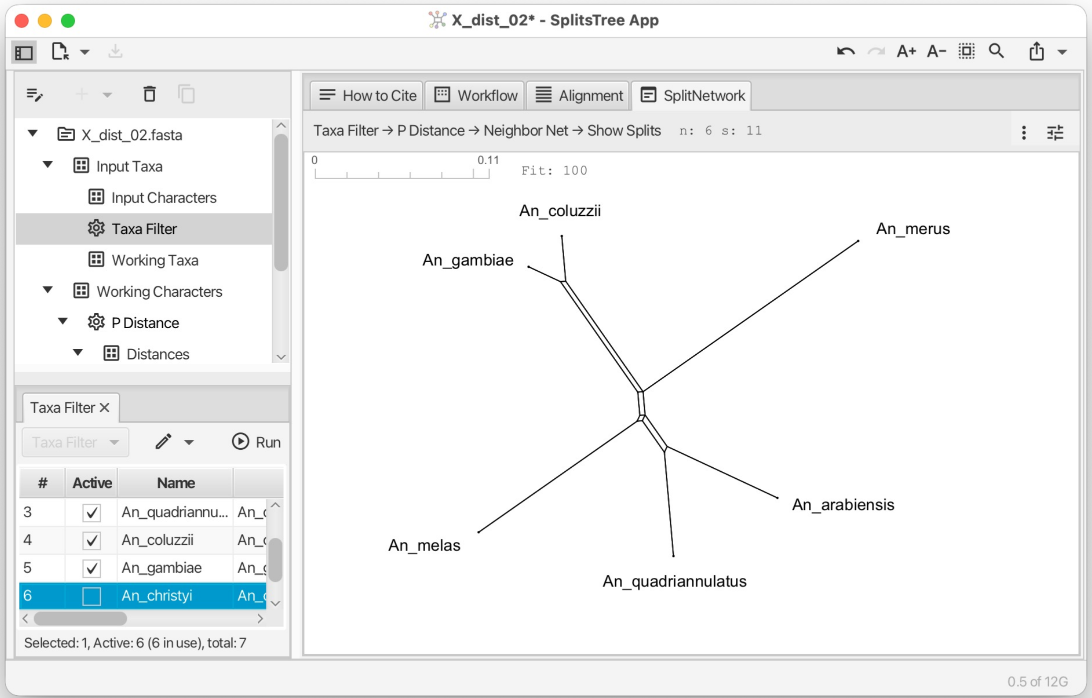
Let's do one together: X_dist_02

- Open file `data/alignments/X_dist_02.fasta` in SplitsTree
- Use the Taxa Filter item (side bar) to deactivate the outgroup:

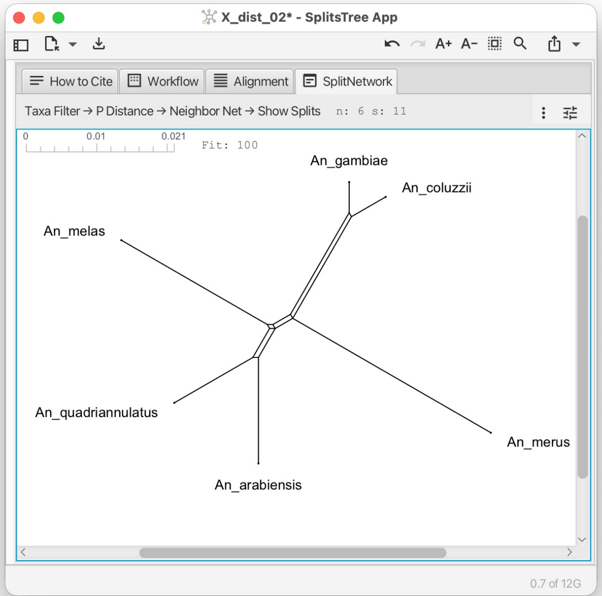


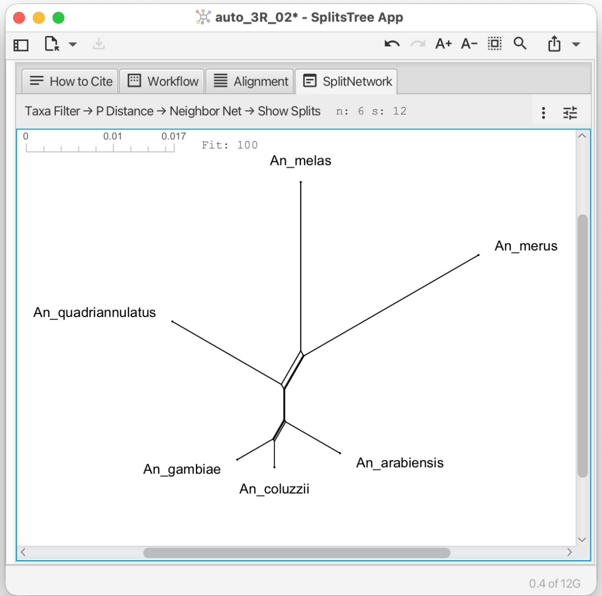
Open other alignments and explore

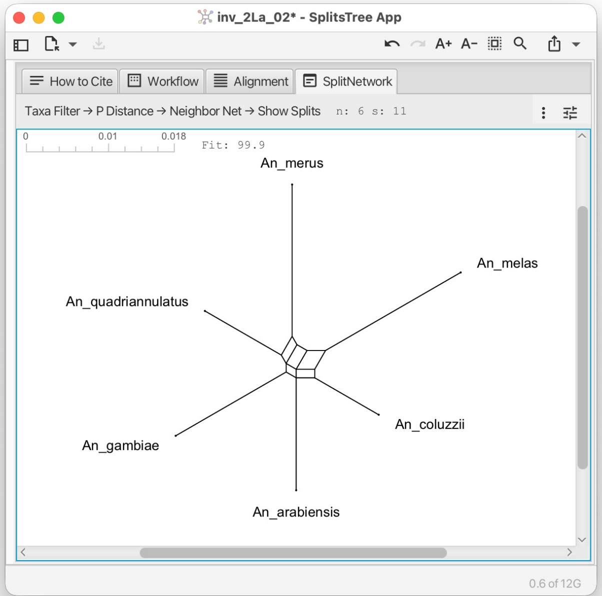
- Repeat the same workflow on alignments from different categories:

Region	Loci	Expected topology
X distal (Xag inversion)	5	(arabiensis, quadriannulatus) - species tree
X pericentromeric	1	(arabiensis, (gambiae, coluzzii)) - introgressed
Autosomal (2R, 3R)	5	(arabiensis, (gambiae, coluzzii)) - introgressed
2La inversion (2L)	2	2La-specific topologies
3La inversion (3L)	2	(merus, quadriannulatus) - unexpected introgression

Open other alignments and explore







- X_dist_02
- "species tree"

- auto_3R_02
- "introgressed"

- Inv_2La_02
- "2La-specific topologies"

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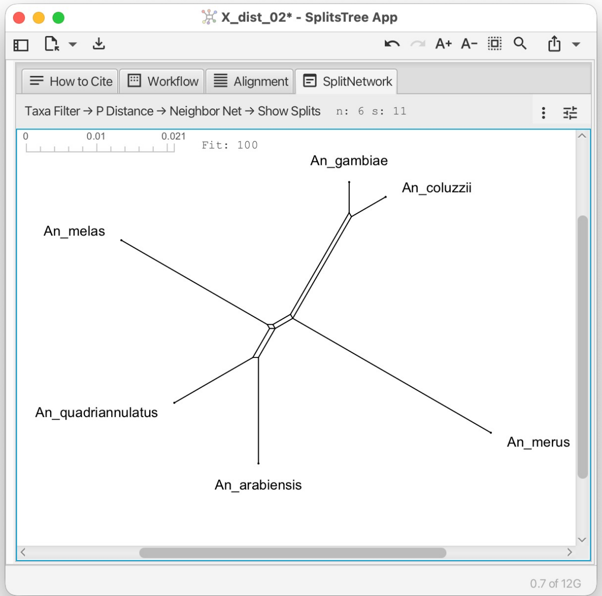
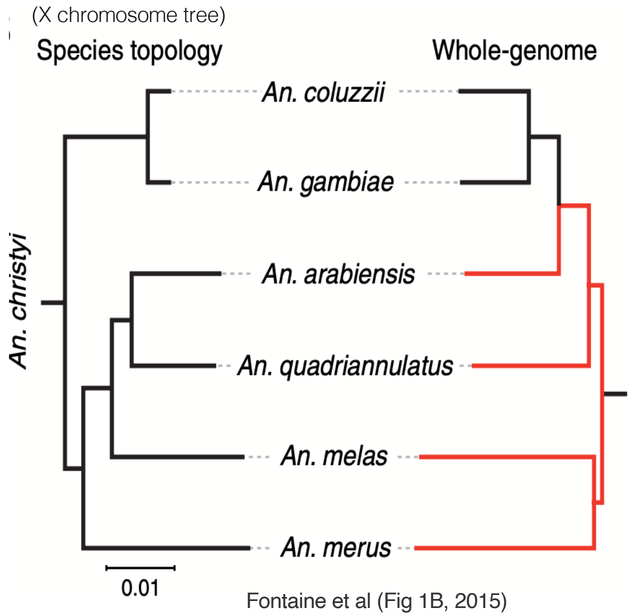
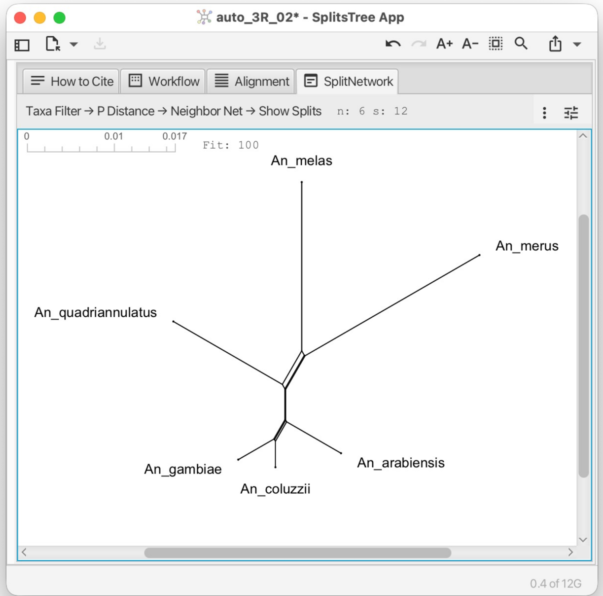
Compare with trees in Fontaine et al (2015)





Fontaine et al (Fig 1B, 2015)



- X_dist_02
- “species tree”

- auto_3R_02
- “introgressed”

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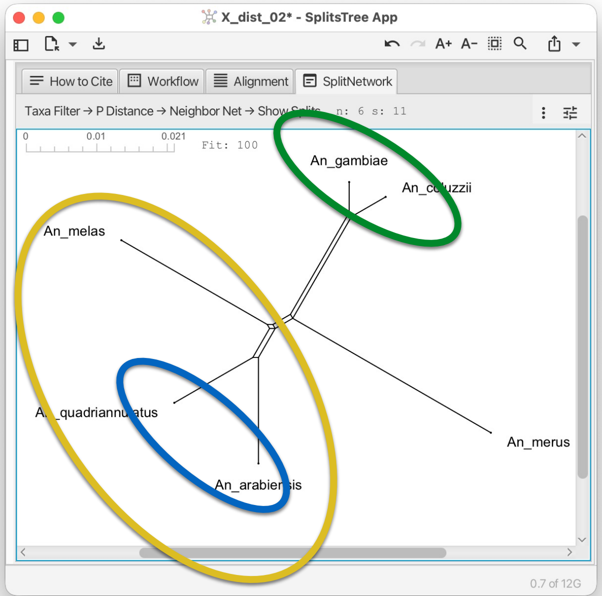
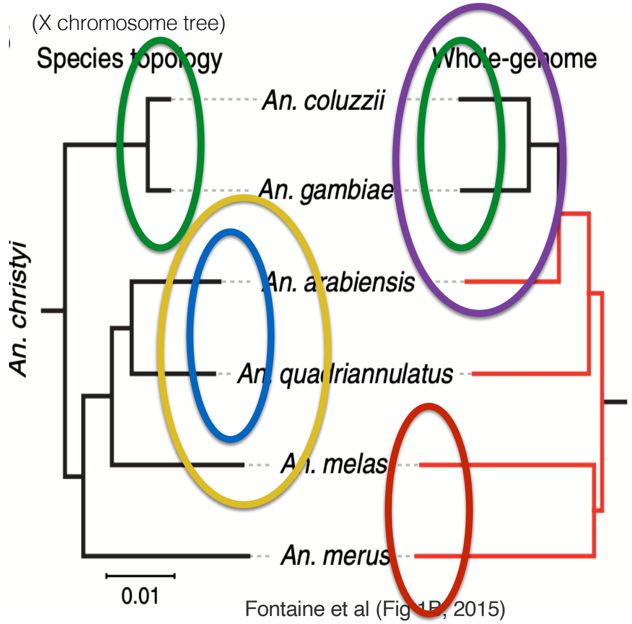
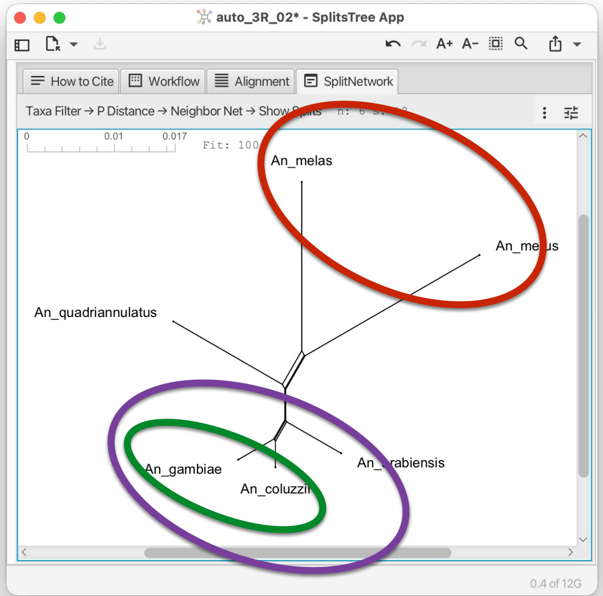
Compare with trees in Fontaine et al (2015)



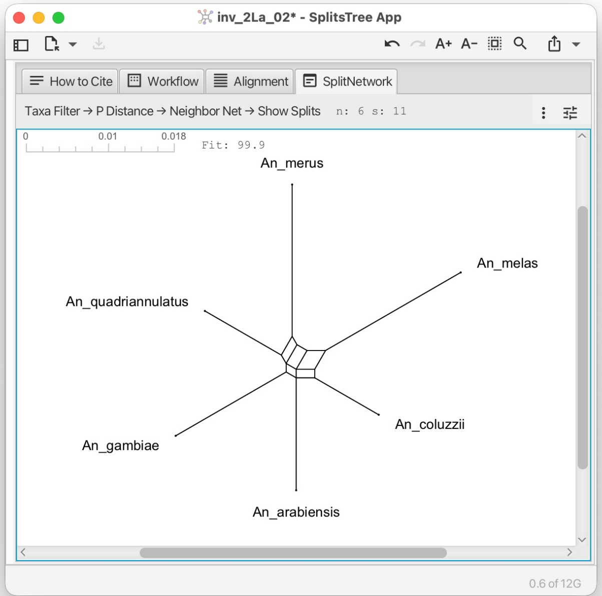


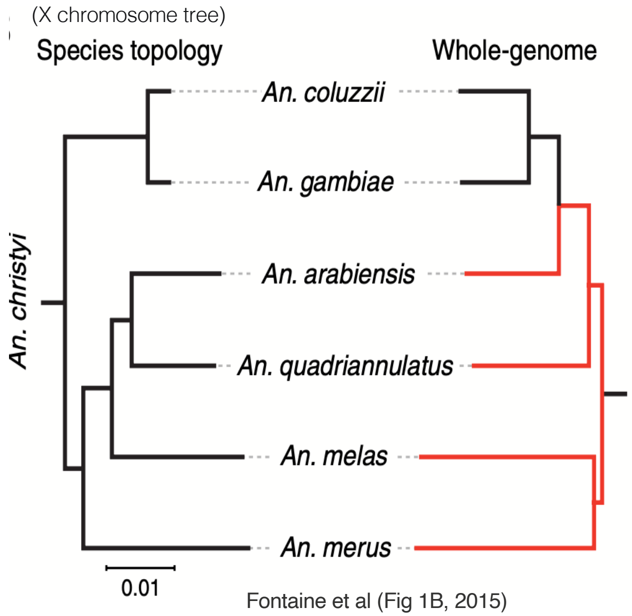


- X_dist_02
- “species tree”

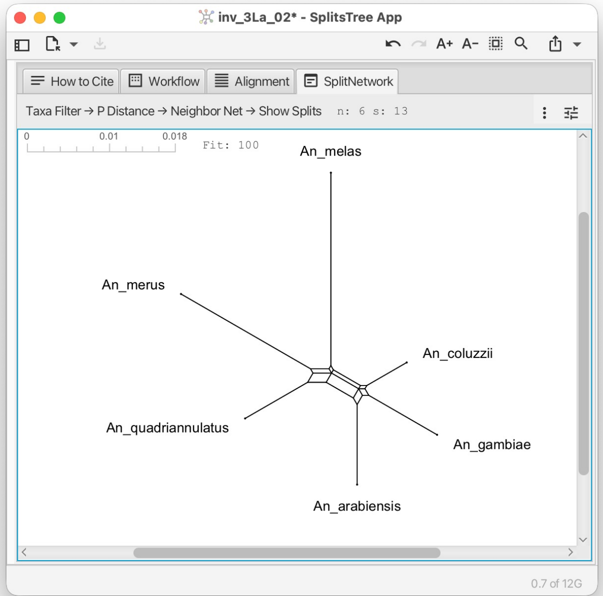
- auto_3R_02
- “introgressed”

Compare with trees in Fontaine et al (2015)





Fontaine et al (Fig 1B, 2015)



- Inv_2La_02
- “2La-specific topologies”

- Inv_3La_02
- “unexpected introgression”

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What did you see?

Across the categories you should have seen:

- X-distal: smaller boxes, closer to tree-like
- Autosomal: more boxes around *An. arabiensis* and (gambiae, coluzzii)
- 2La / 3La inversions: distinctive boxes (3La groups *An. merus* with *An. quadriannulatus*)

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And what’s next?

- Neighbor-Net gives an implicit network — it flags conflict.
- Next we build an explicit network by hand, drawing reticulation nodes for specific hypothesized events.
- This is the “phylogenetic sketching” workflow.

Hands-on: sketch a hypothesis network in PhyloSketch

- Time budget: 20 minutes
- Goal: hand-build a network capturing your best hypothesis for *Anopheles* evolution.
- Tool: PhyloSketch (Huson, 2025)
- Idea: turn the patterns you saw in gene trees and Neighbor-Nets into a biological hypothesis, drawn as an explicit network.
- Scaffold: use Fontaine et al. Fig 1C (next slide) as your target — aim to reproduce its main reticulations.

PhyloSketch in 90 seconds

- Launch PhyloSketch
- Draw: click to place taxa, drag to connect them
- Tree: connect leaves through internal nodes
- Reticulation: convert an internal node, then add a second parent edge
- Edit node label using context menu
- Save: File → Save As → .psketch

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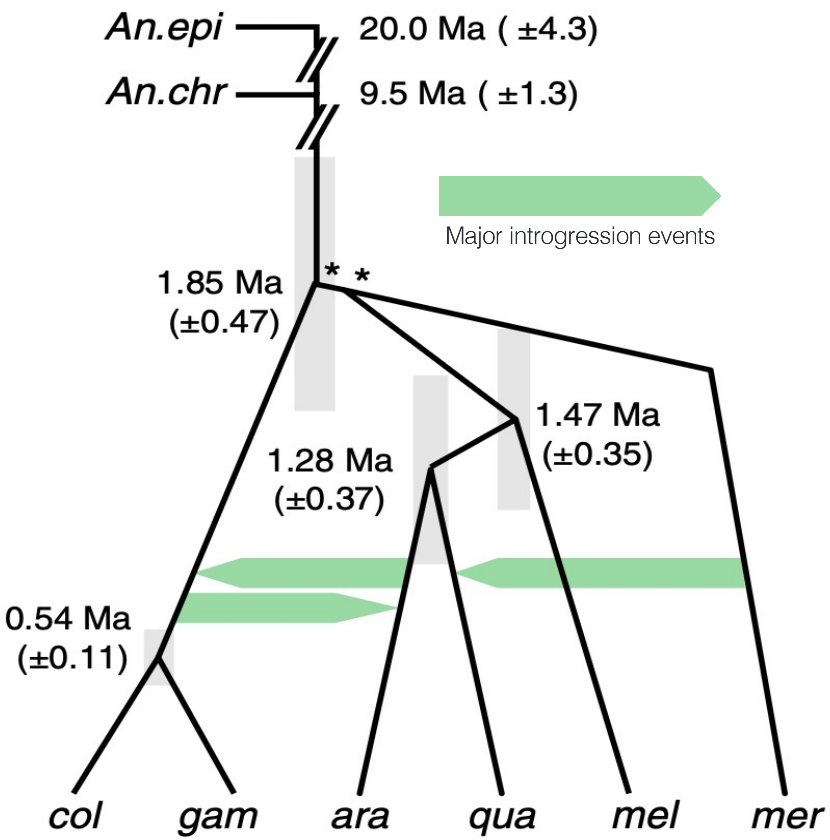
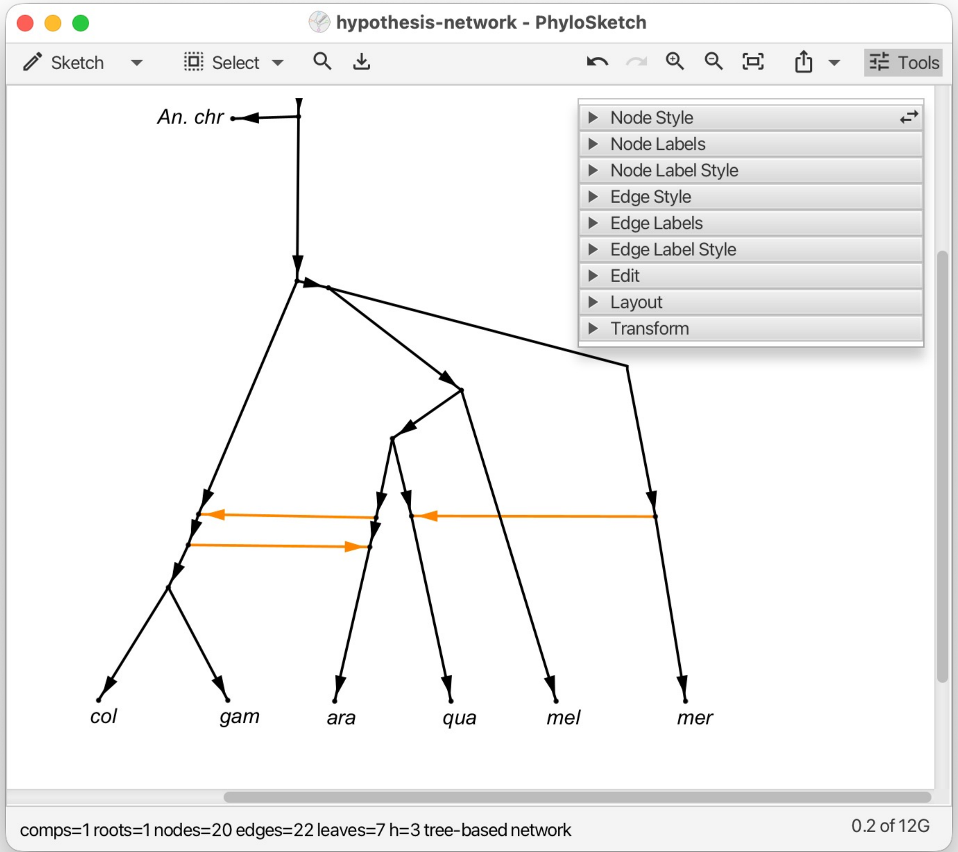
Your target network: Fontaine et al. Fig 1C



Fontaine et al (Fig 1C, 2015)



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Build your hypothesis network

Step 1: Start from the species tree we believe is correct

From X-distal loci, the species tree is:

```
((((arabiensis,quadriannulatus),
(gambiae, coluzzii)),melas),merus);
```

with `christyi` as outgroup. Sketch this in PhyloSketch first.

Step 2: Add reticulation nodes for hypothesized introgression

Events to consider:

Event	Evidence in our data
Introgression from (gambiae, coluzzii) into *An. arabiensis*	Autosomal gene trees show *An. arabiensis* grouped with (gambiae, coluzzii) instead of *An. quadriannulatus*
Introgression between *An. merus* and *An. quadriannulatus*	3La inversion gene trees show (merus, quadriannulatus) sister

Build your hypothesis network

Step 3: Decide which events to include

- A minimal network has 1 reticulation
- A richer network might have 2 or 3
- Trade-off: every reticulation must be justified by evidence

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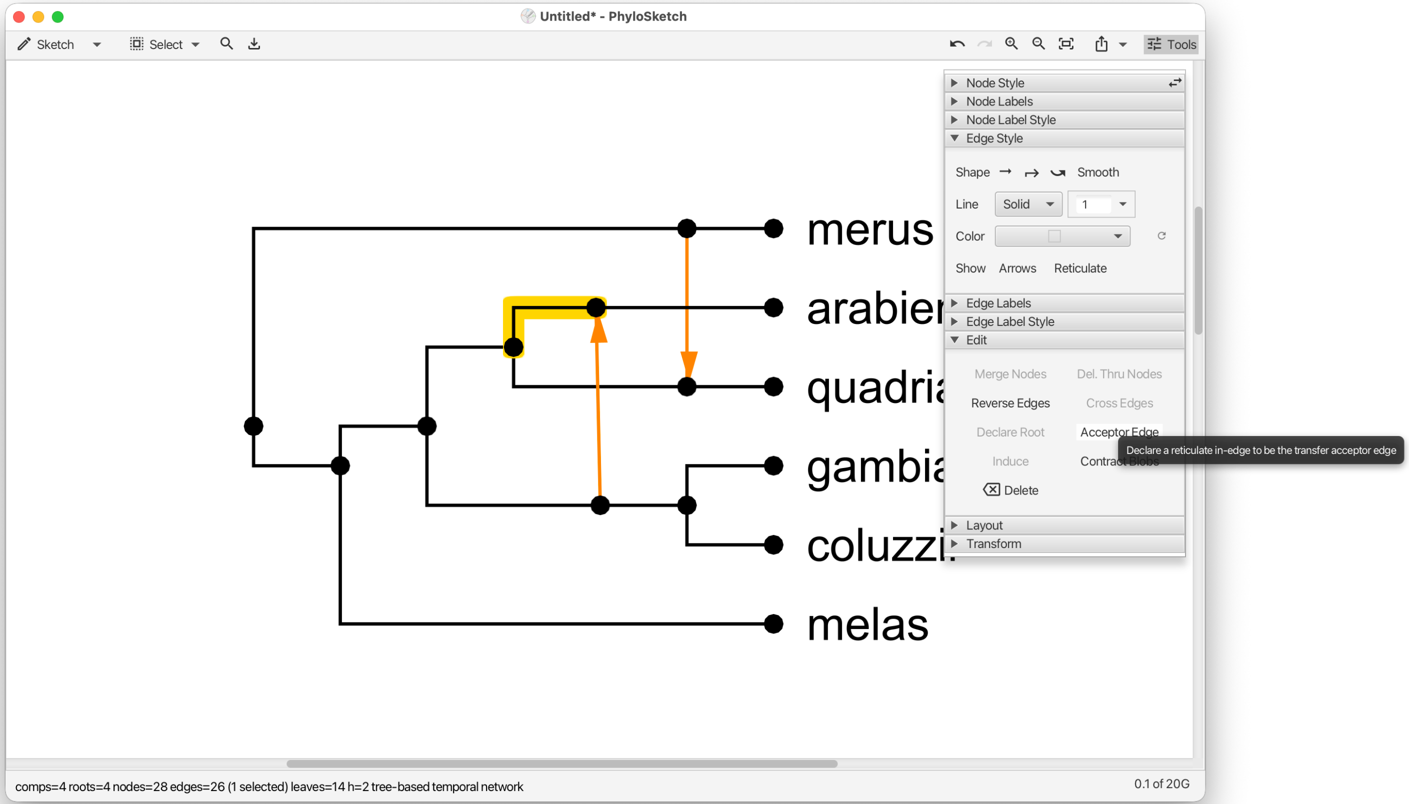
Build your hypothesis network

- To create a transfer edge, declare an edge as “acceptor”:



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10. PhyloParallelograms hands-on
11. Biological interpretation + wrap-up

PhyloGuide Q4

Q: We want to infer an explicit reticulation network from a set of gene trees, with the minimum number of reticulations needed to display all the trees. What method should we use?

A: Use a minimum-hybridization network method, which finds the smallest number of reticulation events explaining all gene trees. Check ILS first because it can mimic reticulation.

Hands-on PhyloParallelograms: understanding gene trees through their network

- Time budget: 30 minutes
- Goal: show your 15 gene trees on an underlying network and see directly where they share history and where they conflict
- Interactively add and remove trees and watch the agreement and conflict change
- Big question: where does the conflict concentrate, and does it match the network you sketched in PhyloSketch?

PhyloParallelograms

By Daniel H. Huson. Contributions: Banu Cetinkaya & Louxin Zhang

(paper submitted)

Hands-on PhyloParallelograms: understanding gene trees through their network

PhyloParallelograms

- A phylogenetic parallelogram draws several rooted gene trees together based on an underlying network
- Where the trees agree, their branches run together as parallel bands; where they conflict, the branches split apart
- The underlying network displays all trees and minimizes reticulations – computed using the PhyloFusion algorithm

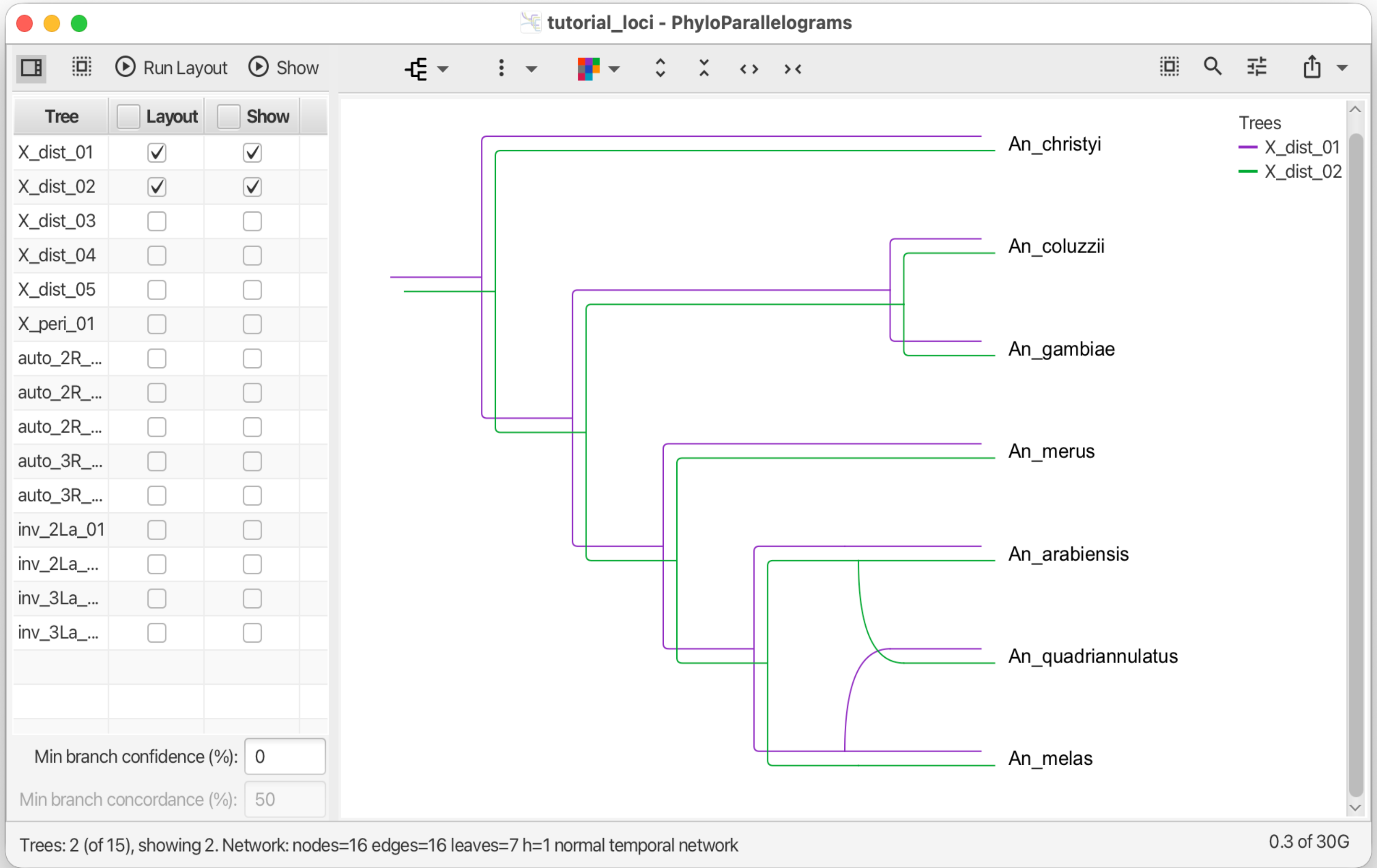
(Zhang, Cetinkaya, Huson 2026)

Hands-on PhyloParallelograms: understanding gene trees through their network

- Launch PhyloParallelograms
- File → Open → tutorial_loci.nex (the 15 gene trees with locus IDs)
- It lists all 15 trees; by default the first two are combined into a network.
- Explore different sets of trees to compare.

Hands-on PhyloParallelograms: understanding gene trees through their network

Two X-distal loci (X_dist_01, X_dist_02). The two trees run together almost everywhere, and the underlying network needs just one reticulation (h=1) — these loci tell nearly the same story, close to the true species history. (h = reticulations in the network.)



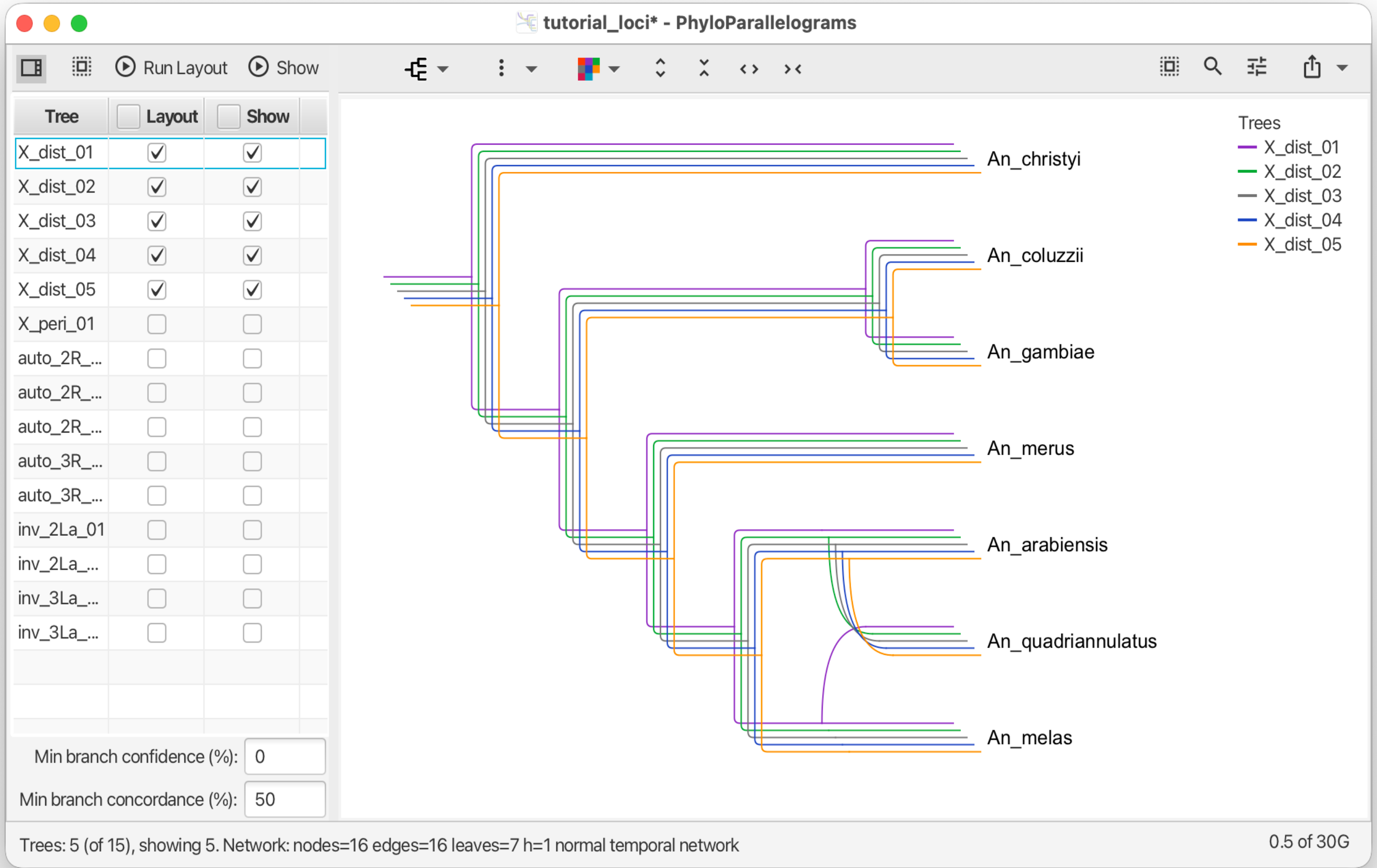
Hands-on PhyloParallelograms: understanding gene trees through their network

The point of the tool is interactive comparison. Try the following subsets and see how the network changes:

- The X-distal loci only (5 trees).
- The inversion loci only (4 trees).
- All 15 trees together.
- Different mixed subsets of trees.

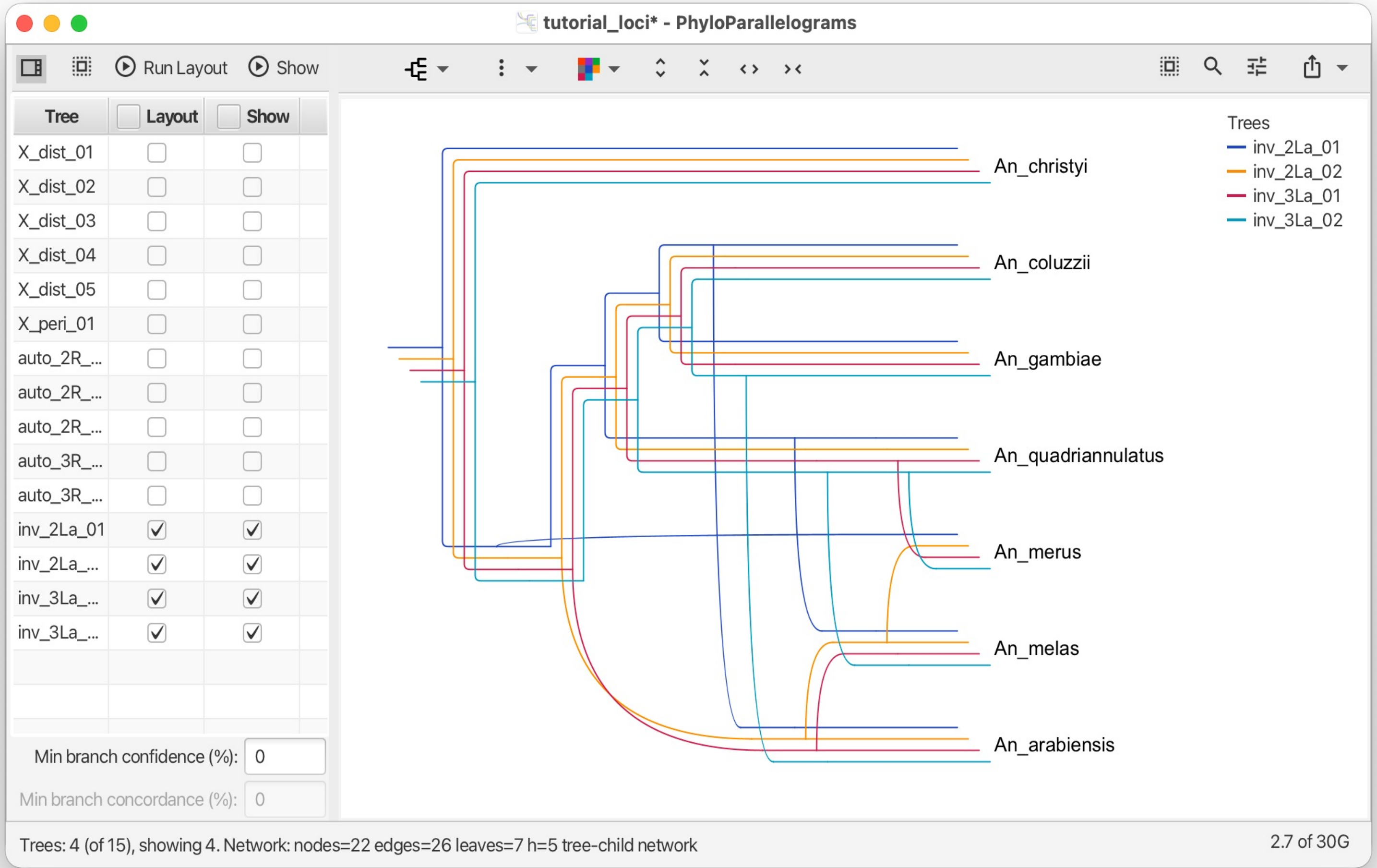
Hands-on PhyloParallelograms: understanding gene trees through their network

All five X-distal loci. Mostly parallel: the X chromosome carries the species branching order (with h=1)



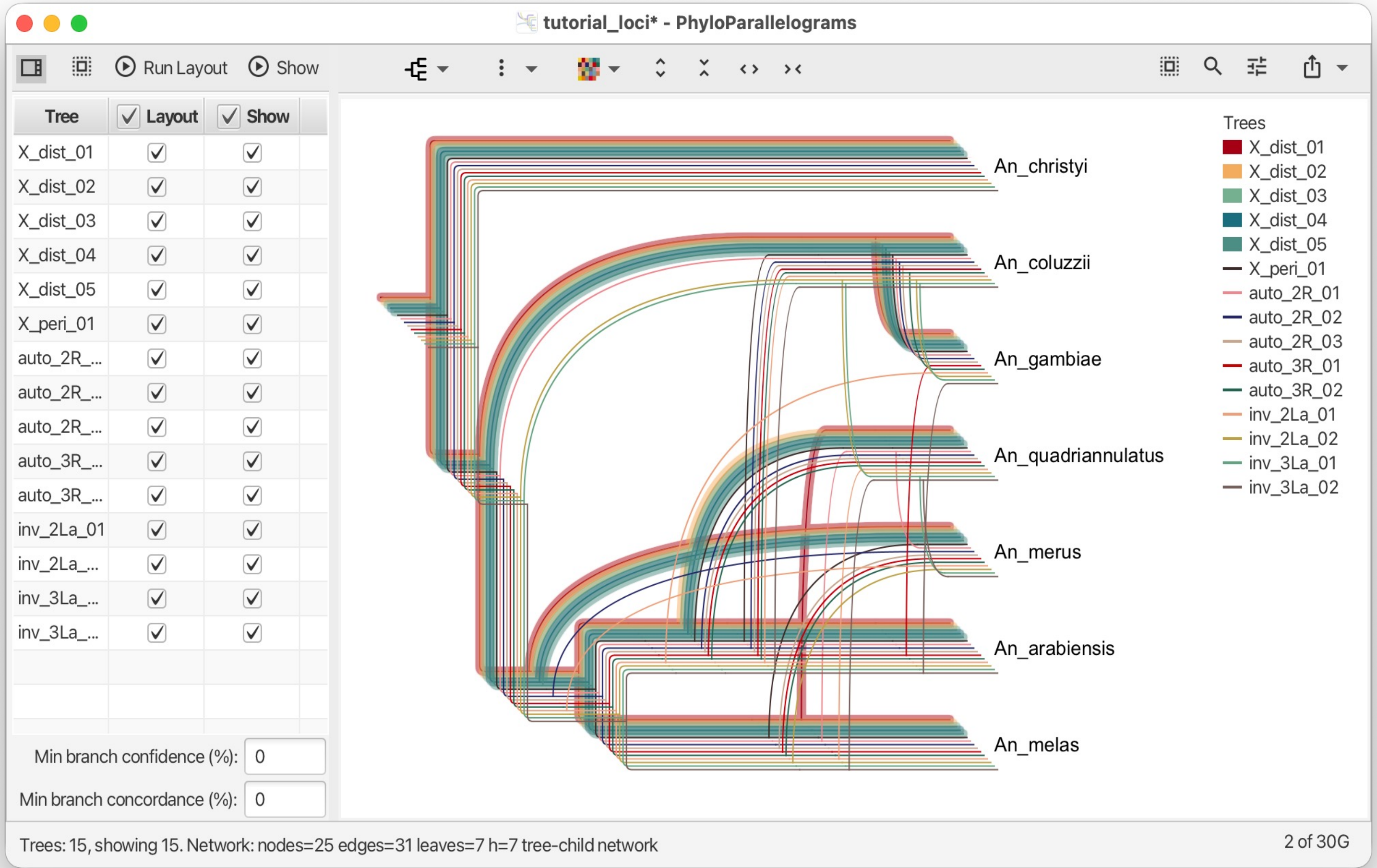
Hands-on PhyloParallelograms: understanding gene trees through their network

Four inversion loci (2La, 3La). The lines now diverge, especially around *An. arabiensis*, *An. quadriannulatus* and *An. merus*, and the network jumps to h=5. Fewer trees than the X set, but far more conflict — these are the introgression paths.



Hands-on PhyloParallelograms: understanding gene trees through their network

All 15 gene trees at once. Maximum divergence across the genome, the underlying network at h=7 — the whole pattern, concordant X against introgressed autosomes and inversions, in a single network.



How does the parallelogram compare to what you sketched?

- Open your PhyloSketch sketch next to the PhyloParallelograms network on all 15 trees.
- Compare: Same number of reticulations? Same events (same lineage pairs)? Same direction of gene flow?

PhyloGuide Q5

Q: We have a phylogenetic network for the Anopheles gambiae species complex with reticulations between An. arabiensis and the (An. gambiae, An. coluzzii) clade, and between An. merus and An. quadriannulatus. What does this tell us biologically?

A: The reticulations suggest non-tree-like evolution, possibly due to historical gene flow/hybridization or ILS. Different loci may reflect different evolutionary histories across the genome.

PhyloGuide

PHYLOGENETICS, GROUNDED

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Agenda

1. Tutorial overview and dataset
2. Setup verification
3. Multi-locus phylogenomics lecture
4. IQ-TREE 3 hands-on
5. BEAST X + SplitsTree posterior visualization
6. ASTRAL hands-on ☕
7. Networks lecture
8. SplitsTree Neighbor-Net hands-on
9. PhyloSketch hands-on
10. PhyloParallelograms hands-on
11. Biological interpretation + wrap-up

Back to the biology: what did this tell us about malaria

- Species branching order: the X-chromosome topology is correct — *An. arabiensis* and *An. quadriannulatus* are sisters, not the three vectors.
- Pervasive autosomal introgression, especially between *An. arabiensis* and the (*An. gambiae*, *An. coluzzii*) ancestor.
- Further introgression between *An. merus* and *An. quadriannulatus* in the 3La region.
- Implication: if introgression moves vector-capability traits, controlling one species may be undercut by introgression from another.

Take-aways from the tutorial

- Gene trees disagree systematically under pervasive introgression — this is data, not noise.
- Tree-summary methods can be confidently wrong: ASTRAL got the wrong species tree because it does not model introgression.
- Networks make the conflict visible: at the alignment level (Neighbor-Net), within-locus (consensus), and across loci (PhyloParallelograms).
- Sketching a hypothesis first makes the automated output interpretable.
- AI assistance (PhyloGuide) helps orient but does not replace expert judgment.

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Thank you - and please give us feedback

- Thanks to:
  - All of you for spending the morning with us
  - The Anopheles genomics consortium and Fontaine et al. for the dataset used here
  - The tool developers whose work underlies the tutorial (Minh lab, Suchard lab, Drummond lab, Rambaut lab, Mirarab lab, and our own group)
  - ISCB for the tutorial program
  - University of Tübingen for support

Share Your Feedback

Scan the QR code to let us know your thoughts on this tutorial.
Thank you!

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